PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – I

INTRODUCTION

Syllabus : Principles of Pre-Stressing – Prestressing Systems – Pre-Tensioning and Post-Tensioning – Advantages and Limitations of Pre-Stressed Concrete – Need for High Strength Materials. Methods of Pre-Stressing : Pre-Tensioning (Hoyer System) and Post-Tensioning (Freyssinet System and Gifford-Udall System).

Contents of this unit

1. Introduction & basic concept of prestressing
2. Principles of prestressing (3 concepts)
3. Types / classification of prestressing
4. Pre-tensioning – procedure
5. Post-tensioning – procedure
6. Pre-tensioning vs Post-tensioning
7. Advantages, limitations & applications
8. PSC vs RCC
9. Need for high strength concrete & steel
10. Hoyer system
11. Freyssinet system
12. Gifford-Udall system (+ Magnel-Blaton, Lee-McCall)
13. Important questions & quick revision

Exam tip : Draw neat sketches for every system (Hoyer, Freyssinet, Gifford-Udall) – 50 % of marks are for the diagram and the sequence of operations.

1. Introduction

Concrete is strong in compression (f_{ck} high) but very weak in tension (≈ 10 % of f_{ck}). So plain concrete beams crack early under load.

- In RCC, steel takes the tension **after** concrete has cracked → cracks are always present, concrete below N.A. is wasted.
- In PSC, a **compressive force is applied in advance** (before external loads) so that tension is cancelled / reduced.

Definition (T.Y. Lin) : Prestressed concrete is concrete in which there have been introduced internal stresses of such magnitude and distribution that the stresses resulting from the given external loadings are counteracted to a desired degree.

Basic idea – the "stack of books" example

Pick up a row of books by pressing them horizontally at the ends – the row acts like a beam and carries its own weight even though there is no tensile strength between books. The squeezing (prestress) makes the compressive force; friction holds them. Similarly a rope of bricks / barrel with iron hoops (circular prestress).



books squeezed by horizontal force P act as a beam

Brief history

- 1886 – P.H. Jackson (USA) – first patent for tightening steel rods in concrete arches.
- 1888 – Doehring (Germany) – prestressed slabs with metal wires. All early attempts **failed** – low-strength steel, losses wiped out the prestress.
- 1928 – **Eugène Freyssinet** (France) – used **high-strength steel** and high-grade concrete; the father of PSC. Shrinkage/creep losses understood.
- 1940s – Magnel (Belgium), Hoyer (Germany) – practical systems. 1950s onwards – bridges, tanks, poles, sleepers.
- India – first PSC bridge near Palghat (Kerala) 1948; IS 1343 is the code of practice (latest 2012).

Key point : Prestress must be of high magnitude, so **high strength steel is a must** – because losses in ordinary steel ≈ prestress itself.

2. Principles of Pre-stressing

Three ways of looking at prestressed concrete (Lin / Guyon):

(i) Prestressing to transform concrete into an elastic material (Stress concept – Freyssinet)

- Concrete = brittle material, poor in tension. Prestressing converts it into an **elastic, un-cracked** material.
- Compressive prestress is applied so that net tension under working loads is **zero or within the permissible limit**.
- Concrete section is analysed as a homogeneous elastic material – ordinary bending theory applies.

$$\text{Concentric : } f = P/A \pm M \cdot y/I$$

$$\text{Eccentric : } f = P/A \pm P \cdot e \cdot y/I \pm M \cdot y/I$$

where P = prestressing force, A = area of section, e = eccentricity, I = moment of inertia, y = distance from C.G., M = external bending moment. (Compression +ve, tension –ve in this note.)

(ii) Prestressing for combination of high-strength steel with concrete (Strength concept)

- PSC is a **combination** of high strength steel and concrete – like RCC.
- Steel takes tension T, concrete takes compression C, forming an **internal resisting couple** of lever arm z.
- Difference from RCC : steel is pre-tensioned, so the whole concrete section is active and the lever arm is larger and almost constant before / after cracking.

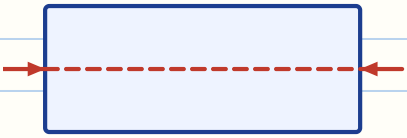
$$M_u = T \times z = C \times z$$

(iii) Prestressing to achieve load balancing (T.Y. Lin)

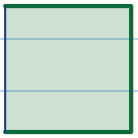
- Prestressing is an attempt to **balance** a part of the external load by the transverse (upward) force from the curved tendon.
- If the net load is balanced, beam has only axial compression (uniform stress) → no deflection, no cracking.

Stress distribution diagrams

(a) Concentric prestress



P through C.G.



uniform P/A

(b) Flexure by loads



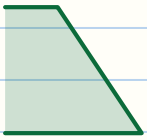
M·y/I : compression top, tension bottom

Resultant : $f = P/A \pm M \cdot y/I$ (zero tension if $P/A \geq M \cdot y/I$)

(c) Eccentric prestress



P at e below C.G.



trapezoid

Self-weight stress + prestress



top $P/A - Pe/Z + M/Z$

bottom $P/A + Pe/Z - M/Z$

Eccentric prestress produces hogging (upward camber) – opposes sagging due to loads.

Camber $\approx P \cdot e \cdot L^2 / (8\ E\ I)$ for parabolic cable with constant P

Load balancing : Parabolic tendon with dip e in span L, force P exerts an **upward UDL** = $8 \cdot P \cdot e / L^2$. For a balanced load w_b:

$$w_b = 8\,P\,e / L^2 \rightarrow P = w_b L^2 / (8\,e)$$

Worked example (load balancing)

A beam of span 10 m carries w = 20 kN/m (total). Tendon parabolic with e = 0.25 m. Find P to balance the load completely.

$$P = w L^2 / (8e) = 20 \times 10^2 / (8 \times 0.25) = 2000/2 = 1000\ kN$$

Under this load the beam is under uniform compression P/A only – no bending, no deflection.

3. Types / Classification of Prestressing

Basis	Types	Remarks
Degree of prestress	**Full** – no tension at all under working load (Type 1) **Limited** – tension within permissible, no cracking (Type 2) **Partial** – cracks allowed but width limited (Type 3)	Type 3 needs additional mild steel (hybrid)
Method of tensioning	**Pre-tensioning** – steel tensioned before concreting **Post-tensioning** – after concrete has hardened	Both are covered in this unit
Bond	**Bonded** – grouted ducts / direct bond **Un-bonded** – greased & sheathed strands	Unbonded: slabs, buildings
Location of tendon	**Internal** – inside concrete **External** – outside the section (box girder, strengthening)	External can be re-stressed
Shape of member	**Linear** – beams, slabs, girders **Circular** – tanks, pipes, silos, domes (wire-wound)	
Tendon profile	Straight, bent (harped), parabolic, circular	
Source of stress	Mechanical (jacks), hydraulic, electrical (heating), chemical (expanding cement)	Hydraulic jacks most common

Prestressing systems – overview

- **Pre-tensioning :** Hoyer system (long line bed, bond transfers stress).
- **Post-tensioning :** Freyssinet (cone & wedge), Gifford-Udall (single wire, cone anchorage), Magnel-Blaton (sandwich plates, wedges), Lee-McCall (threaded bars), CCL, VSL, BBRV (button heads).
- Classification on anchoring : wedge action (Freyssinet, Gifford-Udall, Magnel), direct bearing (button-head BBRV, nuts & threaded bar – Lee-McCall), looped (Baur-Leonhardt, Leoba).

Stages in any system : (1) tensioning → (2) anchoring → (3) transfer of prestress to concrete. Remember the three words **Tension – Anchor – Transfer**.

4. Pre-Tensioning

Steel wires/strands are **tensioned first**, concrete is cast around them, and when concrete attains sufficient strength the tendons are released – stress is transferred to concrete by **bond** (and transmission length).

Procedure (step by step)

1. Long casting bed (50–100 m) with **anchor abutments** at both ends.
2. Tendons are laid along the bed and **tensioned by hydraulic jack** against the abutment; anchored with grips / chucks.
3. Moulds are placed, reinforcement cages (stirrups) fixed, and **concrete is poured** and compacted – tendons remain stretched.
4. Concrete is cured (steam curing speeds up) until it attains **transfer strength ≥ 0.5 – 0.7 f_{ck}** (≈ 3–4 days with steam).
5. Tendons are **gradually released** (jacks / sand boxes / cutting by flame – gradual). Steel tries to shorten → grips the concrete → compression in concrete.

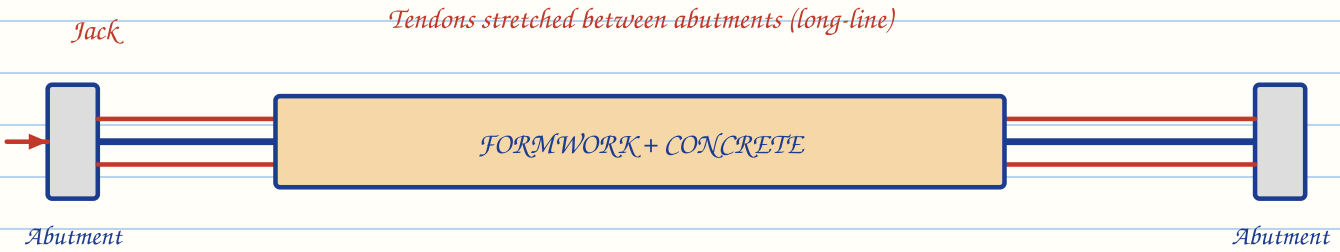


Fig. Pre-tensioning – long line method

Key points : No anchorage device in the member (bond is the anchor) • Tendons are generally straight or harped • Suitable for **factory production** of small / medium precast units – poles, sleepers, planks, beams, pipes • Losses : elastic shortening, shrinkage, creep, relaxation (no friction, no anchorage slip in member).

Transmission length

Length from the end over which the prestress builds up fully by bond. Depends on wire diameter, strength of concrete, surface (indented wires better). IS 1343 : $L_t \approx 30–50\ \varphi$ ... (detailed in Unit III).

5. Post-Tensioning

Concrete is cast first (with **ducts / sheaths** left in it); when it has gained strength the tendons are **stretched against the hardened concrete** using jacks and anchored at the ends. Stress is transferred by **bearing** through end anchorages.

Procedure (step by step)

1. Cast concrete member with a **duct** (corrugated metal / HDPE sheath) placed to the required profile (straight / curved) – with end **anchor blocks** and helix / bursting steel.
2. Cure until concrete reaches the specified **transfer strength** ($\geq 0.7\, f_{ck}$ or as per design, min 25–30 MPa).
3. Thread tendons (wires / strands / bars) through the ducts.
4. **Stress** the tendons with hydraulic jack reacting against the member itself; lock the **anchorage** (wedge / nut / button head). Jack force is held while anchoring.
5. **Grout** the ducts (cement + water + admixture, w/c ≈ 0.40–0.45) – protects steel from corrosion and gives bond (bonded post-tensioning).

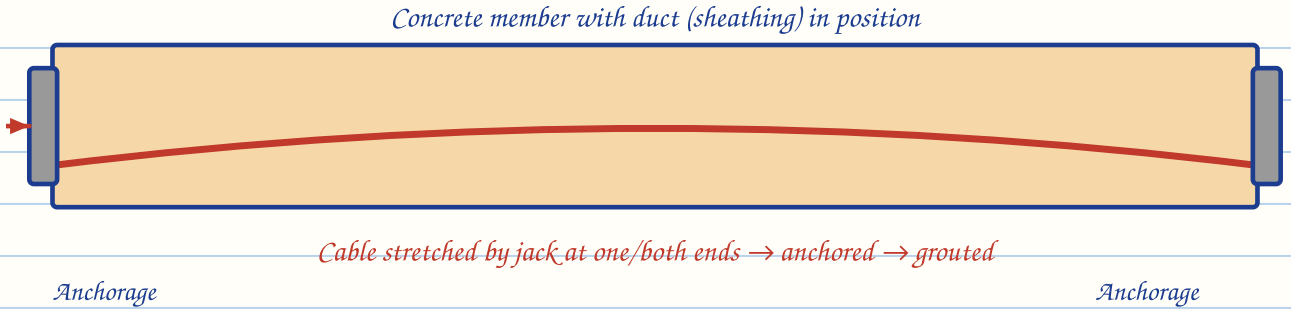


Fig. Post-tensioning schematic

Key points : Anchorage is essential – transfers P by bearing • Curved (parabolic) tendons possible → ideal for long-span bridges, large girders • Done at site (cast in-situ) or on precast units • Losses : elastic shortening (average), shrinkage, creep, relaxation, **friction** and **anchorage slip**.

Remember : Pre-tension = "Tension, then cast". Post-tension = "Cast, then tension".

6. Pre-tensioning vs Post-tensioning

Sl.	Pre-tensioning	Post-tensioning
1	Tendons stretched **before** casting concrete	Tendons stretched **after** concrete hardens
2	Stress transferred by **bond**	Stress transferred by **bearing** (anchorage)
3	No anchorage hardware in member – cheaper per member	Anchorages required – costly
4	Needs long casting beds, abutments – heavy initial investment	Can be done on site; no abutment needed
5	Suitable for small/medium precast mass-produced units	Suitable for large members, long spans, cast-in-situ
6	Straight or harped tendons only (bent at deviators)	Any profile – parabolic, curved easily
7	Eccentricity limited; profile control difficult	Profile controlled by ducts
8	Transmission length at the ends needed	Anchorage zone – end block, bursting steel
9	Losses : elastic shortening (full), creep, shrinkage, relaxation	Also friction + anchorage slip; elastic shortening partly nil if tendons stressed one by one → small
10	Grouting not needed	Grouting of ducts needed (for bonded)
11	Quick production, steam curing, high economy for repetition	Slower, labour intensive
12	Examples : railway sleepers, poles, hollow-core slabs, T-beams	Examples : bridge girders, tanks, containment, long beams

Frequently asked : **"Distinguish between pre-tensioning and post-tensioning" – 5 marks**. Write at least 8 points as above.

7. Advantages of Prestressed Concrete

1. **Section is fully utilised** – whole depth is un-cracked, so I_{gross} is used. Lesser depth, lighter members.
2. **Saving in materials** – 20–35 % concrete and 60–70 % steel (HTS), though steel is costlier per kg.
3. **No cracking** under working loads – better durability, corrosion protection, water tightness (tanks).
4. **Reduced deflection** – camber counteracts load deflection; slender members and long spans (up to 100 m +).
5. **Improved shear resistance** – principal tension reduced by axial compression; web can be thinner.
6. **Better fatigue & impact resistance** – stress range in steel is small.
7. **High-quality control** – the member is tested in effect during stressing (a "proof load").
8. **Quick erection** – precast components, early removal of formwork; less maintenance.
9. **Elastic recovery** – cracks (if any under overload) close on removal of overload.
10. Used for **special structures** – bridges, nuclear containment, tanks, offshore platforms, water towers.

Limitations / Disadvantages

1. Needs **high-quality materials** (HSC & HTS) – costlier, availability.
2. **Skilled labour** and special equipment (jacks, anchors, grouting) required.
3. **Initial cost** (casting bed, anchorage hardware) high – not economical for small spans / low loads.
4. Prestress **losses** must be estimated carefully – loss of effectiveness over time.
5. Needs **greater supervision** and quality control at every stage – a mistake can be disastrous.
6. **Fire resistance** is lower – HTS loses strength above 400 °C; more concrete cover needed.
7. Brittle failure risk if over-prestressed (over-stress / ultimate behaviour).
8. Difficulty in **repair, strengthening, demolition** (stored energy).
9. Anchorage zone needs **special reinforcement** (bursting, spalling).

Applications

Bridges (girders, box, cable-stayed), industrial floors, roofs, silos, tanks, pipes, poles, railway sleepers, piles, dams (anchors), nuclear reactor vessels, offshore towers, runway pavement, retaining walls & rock anchors.

8. Prestressed Concrete vs Reinforced Concrete

Aspect	RCC	PSC
Section	Cracked below N.A.; only part (compression zone) effective	Whole section effective, un-cracked
Steel	Mild steel / HYSD, f_y = 415–550 MPa; stress ≈ 140–230 MPa	HTS, f_{pk} = 1570–1860 MPa; working stress up to ≈ 1000+ MPa
Concrete	M20 – M30 typical	M35 – M60 (higher)
Deflection	Large; creep deflection more	Small (camber)
Span / depth	L/d = 20–26	L/d = 25–35, longer spans
Cracks	Always present at service	Not present (full prestress)
Self-weight	Heavy, deeper sections	Light, slender sections
Stress state	Steel stress varies with loads	Steel stress nearly constant
Shear	Needs more stirrups	Reduced by prestress
Skill/Cost	Easy; low cost for short span	Skilled; economical for long span

9. Need for High Strength Materials

(a) High strength concrete

- **High compressive strength** is needed to resist the high stresses at transfer – stress concentration near anchorages (bearing stresses).
- Smaller creep & shrinkage strain (higher E_c) → **smaller losses**.
- Higher **bond strength** (pre-tension) and **shear** strength.
- Higher tensile strength → resists cracking (modulus of rupture $f_{cr} = 0.7\sqrt{f_{ck}}$).
- Economy – smaller sections, lighter members, faster (early) strength gain permitting early transfer.
- IS 1343 : min M30 for post-tensioned, M40 for pre-tensioned members.

(b) High strength steel – why mild steel will not do

Consider a 6 m long member. Initial prestress in the steel must be large because **losses are fixed strain losses** (shrinkage + creep + elastic ≈ 0.0007–0.0015 m/m in total).

Example : Total loss strain in concrete (shrinkage + creep + elastic) ≈ **0.0008** (typical). Steel $E_s = 2 \times 10^5$ N/mm².

$$\text{Loss of stress} = \varepsilon \times E_s = 0.0008 \times 2\times10^5 = 160 \text{ N/mm}^2$$

Steel	Initial stress	Loss	Remaining	% loss
Mild steel	140 N/mm²	160	**–20 (nil!)**	> 100 %
HTS wire	1200 N/mm²	160 (+relax.)	1040 N/mm²	≈ 13 %

So with mild steel the prestress is completely lost; with HTS the effective prestress is retained.

Hence high initial stress → high-strength steel.

Requirements of prestressing steel

- High tensile strength (1570–2000 N/mm²) and high yield (proof stress 0.2 % ≥ 0.85 f_{pk}).
- Adequate **ductility** (min. elongation ≥ 3.5 % on gauge length 200 mm).
- Low **relaxation** (low-relaxation strands ≈ 2.5 % at 1000 h).
- Good bond / surface (indented, crimped, stranded), freedom from corrosion, and ability to be anchored.
- Straightness – coiled wire diameter large enough to avoid residual curvature.

Types of prestressing steel (IS 1785, IS 6006, IS 2090)

Type	Details	f_{pk} (N/mm²)
Cold-drawn wires (IS 1785)	Ø 2.5, 3, 4, 5, 7, 8 mm; plain or indented	1570–1860 (smaller dia → higher)
Strands (IS 6006)	7-wire standard / super; 2, 3 wires also; Ø 9.5, 12.7, 15.2 mm	1770–1860
High-tensile alloy bars (IS 2090)	Ø 10–32 mm, threaded; used with nuts (Lee-McCall)	980–1030

Remember : Wire Ø 7 mm → f_{pk} = 1570 MPa; strand 12.7 mm → 1860 MPa (IS 6006). Modulus $E_s = 2.0 \times 10^5$ N/mm² (wires, bars) and 1.95×10^5 (strands).

10. Methods of Pre-stressing

A. Pre-tensioning : HOYER SYSTEM (Long-line system)

Developed by **E. Hoyer (Germany)** – also called the Hoyer long-line system / "Hoyer effect".

- Piano wires (Ø 2–8 mm) are tensioned between two **massive abutments** 100 m or more apart on a **long casting bed**.
- Many members (e.g. 10–20 sleepers, joists) are cast in line on the same wires, separated by **bulkheads / separators**.
- After concrete has hardened, the wires are **released gradually** by **jacks / sand jacks** at the tension end – so wire tries to regain length, swells in diameter at the cut end (Hoyer effect) and gives a wedge-action bond.
- The wires are then **cut between the members** – each member is separated.

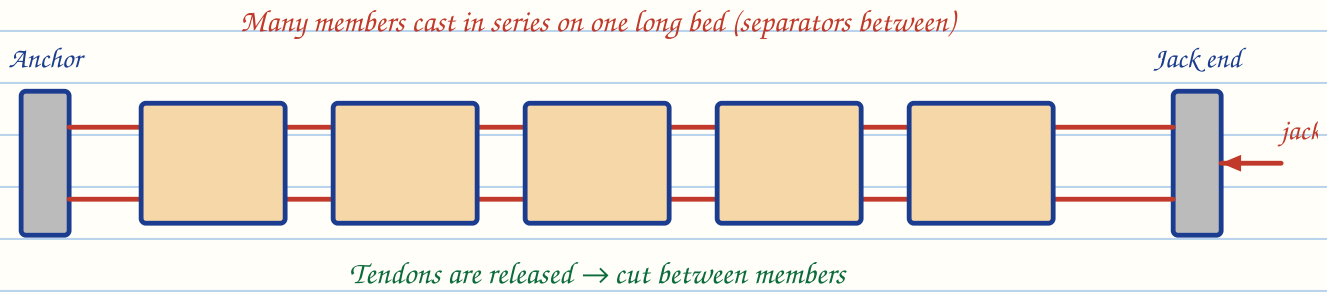


Fig. Hoyer long-line pre-tensioning system

Hoyer effect

When tension is released the wire contracts laterally (Poisson effect) except inside the concrete – the diameter increases back inside the member, producing **wedge action** that improves the bond at the ends and aids stress transfer.

Advantages : economical for mass production, no end anchorage, quick re-use of bed, steam curing.

Limitations : long bed, loss in length of wire between members (waste), only straight tendons, release must be gradual to avoid shock.

B. Post-tensioning : FREYSSINET SYSTEM

Developed by **Eugène Freyssinet (France)**. Wire / strand tendons with **cone and wedge anchorage**.

- Tendon = **12 wires Ø 5 mm or Ø 7 mm** (or 12 strands of 12.7 / 15.2 mm in later versions) placed in a flexible metal duct (sheath).
- Anchorage : a **concrete (or steel) cylindrical female cone** embedded in the end of the member + a **grooved male (wedge) cone** – wires lie in the grooves.
- Jacking : a **double-acting hydraulic jack** pulls the wires; after the required force is reached, the male cone is **forced (pressed) into the female cone** by the 2nd (inner) ram – wedge action grips wires.
- Jack released → wires try to shorten but the wedge grips → prestress transferred by bearing at the anchor.
- Ducts **grouted** under pressure (cement grout) – bond + corrosion protection.

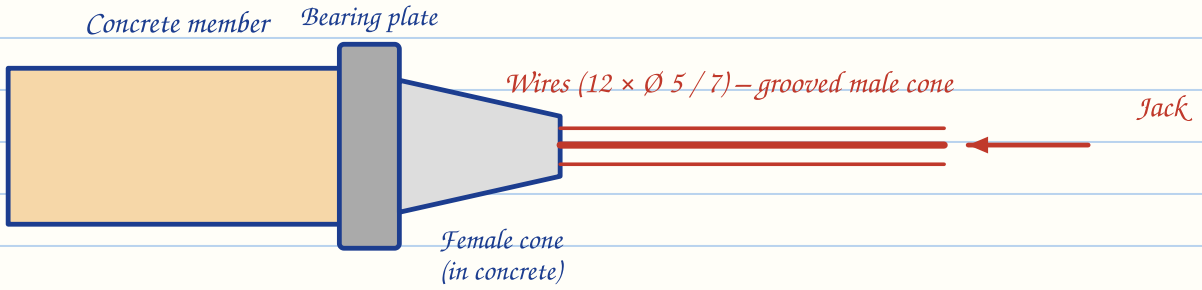


Fig. Freyssinet cone anchorage – principle

Salient features

- Cone angle small (≈ 12°) → good wedge grip. Duct ≈ 28–40 mm Ø. Each tendon takes ≈ 400–1000 kN.
- Wedge-type, **tension at one or both ends**; many tendons can be placed in a section – used for bridges, tanks, silos.
- Anchor slip (wedge draw-in) ≈ 6 mm – to be accounted in losses (Unit II).
- Dead end anchorage, "Freyssinet jack" is of 2 stages – pulling and wedging.

Sequence : Cast – Insert tendons – Fit jack – Stress – Push wedge – Release – Grout.

C. Post-tensioning : GIFFORD-UDALL SYSTEM

British system (Gifford & Udall, 1939) – a **single-wire system** using a very simple **cone grip** anchorage; the **Gifford-Udall-CCL** variant is widely used.

- Each wire (Ø 2–8 mm, usually 5–7 mm) is **stressed individually** by a small **single-wire jack** (CCL jack).
- The wire is passed through an **anchor plate (bearing)** and a split **conical wedge (grip)** – wedges are **pushed in by the jack** after stretching.
- Wires are placed in ducts in groups; in strand version (CCL) 7-wire strands with split-barrel anchors are used.
- After stressing, **wedges lock** the wire; excess length cut off; ducts grouted.

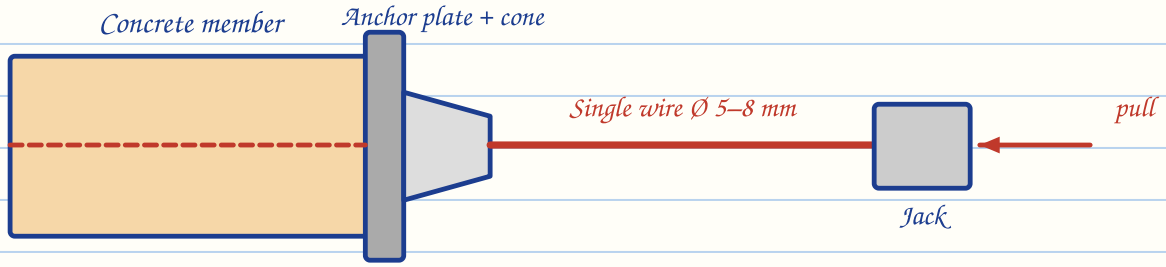


Fig. Gifford-Udall anchorage – cone grip

Features

- Simple equipment, easily portable – used for **small / medium** members as well as large.
- Wires stressed **one by one** → elastic shortening loss is smaller (pre-stress in the first wires is reduced by later wires but can be compensated).
- Anchorage is cheap; loss of prestress by **slip ≈ 1–2 mm**(wedge draw-in) to be allowed for.
- Not suitable if all wires need to be stressed simultaneously (large tendons – use Freyssinet).

Other post-tensioning systems (short notes)

- **Magnel-Blaton (Belgium)** – flat sandwich plates with 2 wires per wedge; 4-wires per layer, 8 wires/tendon; uses steel wedges between sandwich plates – two wires are stressed simultaneously.
- **Lee-McCall** – high-tensile alloy steel **bars** with threaded ends and **nuts** – direct bearing anchorage; low loss (no slip).
- **BBRV** – button-heads formed on wires, bearing on the anchor plate. **VSL** – strands with wedge anchor.

11. Important Questions & Quick Revision

Short answers (2 marks)

1. Define prestressed concrete.
2. State the three concepts of prestressing.
3. Why is high strength steel necessary for prestressing?
4. What is the Hoyer effect?
5. List the differences between bonded and un-bonded tendons.
6. What is transmission length?
7. Name any two post-tensioning systems.

Long answers (10 marks)

1. Explain the basic principles of prestressing with neat sketches. Compare it with RCC.
2. Distinguish pre- and post-tensioning. State the merits and demerits of PSC.
3. Explain the Hoyer system of pre-tensioning with a neat sketch.
4. Describe the Freyssinet system of post-tensioning.
5. Explain the Gifford-Udall system. How does it differ from Freyssinet system?
6. Why do we need high-grade concrete and HTS in prestressing? Give an example.

Quick revision formulae & facts

Item	Value / formula
Stress at fibre (eccentric P)	$f = P/A \pm P\cdot e\cdot y/I \pm M\cdot y/I$
Load balancing (parabolic)	$w_b = 8Pe / L^2$
Min grade (IS 1343)	Post-tension M30, pre-tension M40
Loss strain (typical)	$\varepsilon \approx 0.0008 \rightarrow 160\ N/mm^2$ for $E_s = 2\times10^5$
Transfer strength	$\geq 0.5–0.7\ f_{ck}$
Stage sequence	Tension → Anchor → Transfer
Pre-tension	Bond; Hoyer; long-line
Post-tension	Bearing; Freyssinet (12 wires cone), Gifford-Udall (single wire jack), Magnel (sandwich), Lee-McCall (bars)

Best of luck ! Practise the diagrams – they carry the marks.

PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – II

LOSSES OF PRE-STRESS

Syllabus : Loss of Pre-Stress in Pre-Tensioned and Post-Tensioned Members due to Elastic Shortening, Shrinkage and Creep of Concrete, Relaxation of Stress in Steel, Anchorage Slip and Frictional Losses.

Contents

1. Introduction – types of losses
2. Elastic shortening of concrete
3. Shrinkage of concrete
4. Creep of concrete
5. Relaxation of steel
6. Anchorage slip
7. Frictional losses
8. Total losses – summary table (IS 1343)
9. Worked example – pre-tensioned beam
10. Worked example – post-tensioned beam
11. Practice problems & important questions

Exam tip : Learn the formula for each loss with the symbols, plus one numerical per loss. The full-loss numerical (10 marks) is guaranteed.

1. Introduction

The stress in the tendon immediately after jacking is not the same as the stress that remains in service. **The reduction of the initial prestressing force is called "loss of prestress".**

- Initial prestress P_i (at jacking) → P_0 (immediately after transfer) → P_e (effective, after all losses).
- Effective prestress $P_e = R \times P_i$ where **R = loss ratio / residual factor** (≈ 0.75 – 0.85).
- Losses are due to the shortening of concrete and the shortening / relaxation of steel; **losses can not be avoided** but can be estimated.
- Why estimate? – an **under-estimate** leads to cracking, an over-estimate leads to over-design.

$$\text{Loss \%} = (P_i - P_e)/P_i \times 100$$

Classification of losses

Pre-tensioned members	Post-tensioned members
1. Elastic shortening of concrete	1. Elastic shortening (partial – depends on sequence)
2. Shrinkage of concrete	2. Shrinkage of concrete
3. Creep of concrete	3. Creep of concrete
4. Relaxation of steel	4. Relaxation of steel
(no friction, no slip)	5. **Anchorage slip** 6. **Frictional loss** (length & curvature)

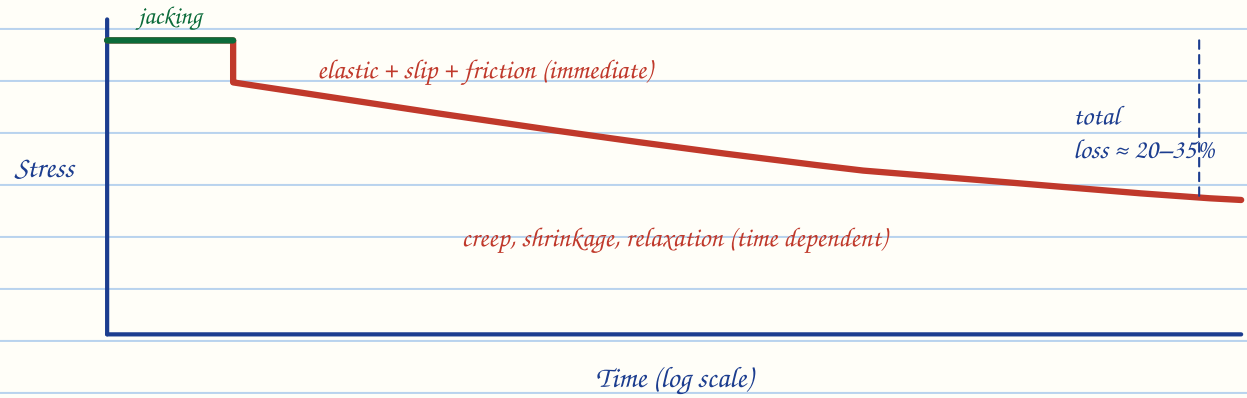


Fig. Variation of tendon stress with time (loss of prestress)

Immediate losses = elastic shortening, friction, anchorage slip. **Time-dependent losses** = creep, shrinkage, relaxation.

2. Elastic Shortening of Concrete

When prestress is transferred, concrete shortens elastically; the steel bonded to it (or anchored) shortens by the same strain → loss of stress in steel.

(a) Pre-tensioned members

- All tendons are released together – concrete shortens by the strain ε_c at the level of steel. The **steel stress drops by the same strain × E_s.**

$$\text{Loss} = \varepsilon_c E_s = (f_c/E_c) E_s = m f_c$$

- $m = E_s/E_c$ = modular ratio; $E_c = 5700\sqrt{f_{ck}}$ (IS 1343) N/mm²
- f_c = stress in concrete at the level of steel due to prestress (and self weight if effective):

$$f_c = P/A + P e^2/I \; (- M_g e/I)$$

where P = prestressing force after transfer ≈ P_i (approx.), A = area, I = M.I. of the section, e = eccentricity, M_g = moment due to self-weight (helps; reduces the stress at steel level – often neglected/taken).

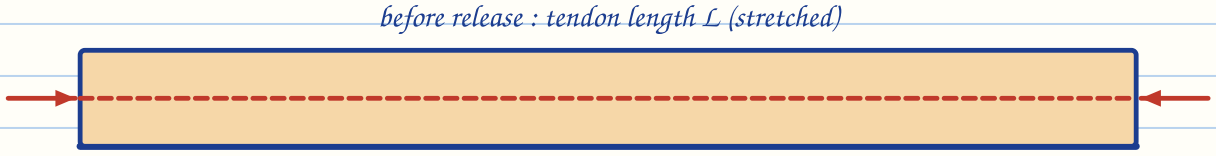


Fig. Elastic shortening – pre-tensioned member

Example 1

A pre-tensioned beam 100 mm × 250 mm is prestressed by 8 wires Ø 7 mm at a constant eccentricity of 50 mm, initial stress 1200 N/mm². $m = 6$. Find the loss due to elastic shortening.

$A_{ps} = 8 \times 38.5 = 308$ mm²; $P = 1200 \times 308 = 369.6$ kN; $A = 25000$ mm²; $I = 100 \times 250^3/12 = 1.302 \times 10^8$

$f_c = 369600/25000 + 369600 \times 50^2/1.302 \times 10^8 = 14.78 + 7.10 = 21.9$ N/mm²

Loss = $m f_c = 6 \times 21.9 =$ **131.3 N/mm²** (≈ 10.9 % of 1200)

(b) Post-tensioned members

- Tendons are stressed against hardened concrete – if **only one tendon** is stressed, the shortening occurs during jacking and is automatically compensated – **no loss**.
- If **several tendons** are stressed one after another, each subsequent tendon shortens the concrete and causes a loss in the previously stressed tendons. The **last tendon has no loss; the first one has the maximum**.

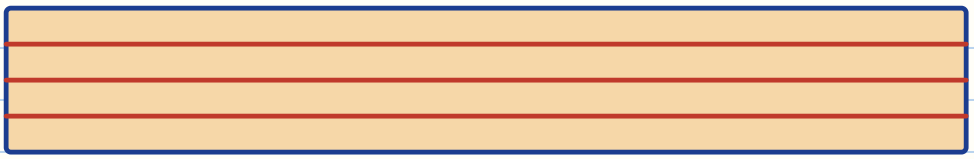


Fig. Sequential stressing of tendons

Let n = number of tendons, stressed successively. Stress in concrete at C.G. of steel due to all tendons = f_c:

- Loss in 1st tendon = $(n-1)/n \times m f_c$; last = 0.
- Average loss over all tendons (what we use):

$$\text{Loss} = \tfrac{1}{2} \cdot (n-1)/n \cdot m f_c$$

For $n = 1 \to 0$; for large $n \to \tfrac{1}{2}\, m f_c$. (Using $m f_c/2$ is a conservative approximation.)

Example 2

A post-tensioned beam, section 150 mm × 300 mm, has 4 cables stressed one after other, each with A_{ps} = 100 mm² at initial stress 1000 N/mm², constant e = 60 mm. $E_s = 2.1\times10^5$, $E_c = 3.5 \times 10^4$. Find the average loss.

$$P = 4 \times 100 \times 1000 = 400\ \text{kN};\ A = 45000;\ I = 150\times300^3/12 = 3.375\times10^8\ \text{mm}^4$$

$$f_c = 400000/45000 + 400000\times60^2/3.375\times10^8 = 8.89 + 4.27 = 13.16\ \text{N/mm}^2$$

$$m = 2.1\times10^5/3.5\times10^4 = 6.0;\ n = 4$$

$$\text{Loss} = \tfrac{1}{2}(3/4) \times 6 \times 13.16 = 29.6\ \text{N/mm}^2\ (\approx 3\ \%)$$

IS 1343 note : For pre-tension members elastic loss is based on the stress at transfer (steel level); for post-tension with several cables, loss may be calculated as above.

3. Shrinkage of Concrete

Shrinkage is the **contraction of concrete due to loss of moisture** (drying) and chemical changes – independent of stress. Steel shortens along with it → loss.

- Factors : w/c ratio, cement content & type, aggregate, humidity, size of member, age at transfer, curing. Higher w/c ratio → more shrinkage.
- **Occurs in both** pre- and post-tensioned members; the time-dependent shrinkage continues for years (~ 80 % in the first year).

IS 1343 values of residual shrinkage strain ε_{cs}

Type	Total shrinkage strain ε_{cs}
Pre-tensioning	0.0003
Post-tensioning	$0.0002 / \log_{10}(t + 2)$ (t = age at transfer, in days)

Loss due to shrinkage = $\varepsilon_{cs} \times E_s$

Post-tensioning loss is smaller because a part of shrinkage has already occurred before stressing (t = age of concrete at transfer). Older concrete → lower loss.

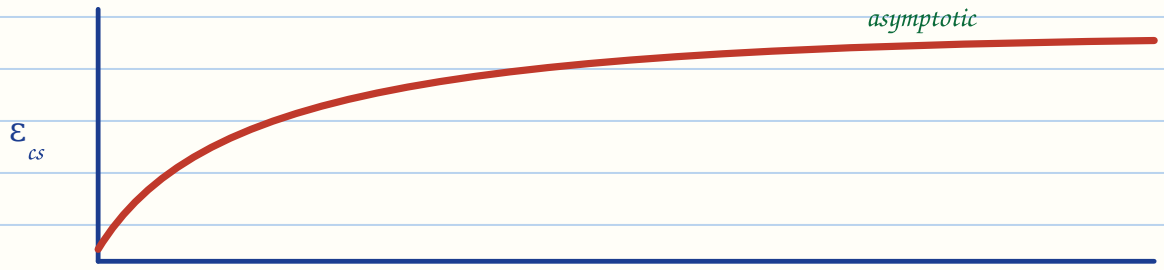


Fig. Shrinkage strain vs age – increases rapidly early, then flattens

Example 3

Calculate the shrinkage loss of (a) pre-tensioned member (b) post-tensioned member with age 10 days at transfer. $E_s = 2.1 \times 10^5$ N/mm².

(a) Pre : $0.0003 \times 2.1 \times 10^5 = 63$ N/mm²

(b) Post : $t = 10$; $\log_{10}(12) = 1.079$; $\varepsilon_{cs} = 0.0002/1.079 = 1.853 \times 10^{-4}$

Loss = $1.853 \times 10^{-4} \times 2.1 \times 10^5 = 38.9$ N/mm²

For the same steel, shrinkage loss in pre-tensioned = 63 N/mm² vs 30–40 N/mm² in post-tensioned (hence higher).

4. Creep of Concrete

Creep is the **time-dependent increase in strain under sustained stress**. Concrete under prestress continues to shorten with time → steel shortens → loss of prestress.

- Depends on : stress level (≈ proportional up to 0.4–0.5 f_{ck}), age at loading (earlier → more), humidity, w/c, aggregate, cement, size, duration.
- Creep strain $\varepsilon_{cc} = \theta \times (f_c / E_c)$ where θ = **creep coefficient** (ratio of creep strain to elastic strain).

Age at loading (transfer)	Creep coefficient θ (IS 1343)
7 days	2.2
28 days	1.6
1 year	1.1

$$\text{Loss due to creep} = \varepsilon_{cc} E_s = \theta \cdot (f_c / E_c) \cdot E_s = \theta \cdot m \cdot f_c$$

- f_c = stress in concrete at the level of centroid of steel (long-term sustained prestress – including self-weight effect).
- Applies to **both pre-tensioned and post-tensioned** members. Interpolate θ for intermediate ages.

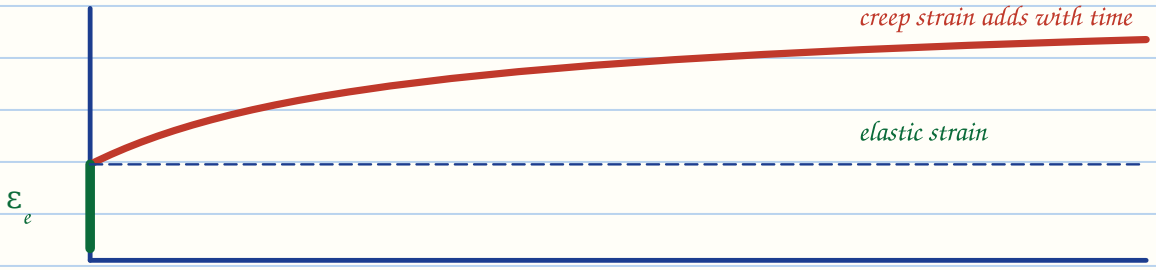


Fig. Creep strain under sustained stress

Example 4

A pre-tensioned beam has f_c = 7.5 N/mm² at the level of steel, transferred at 28 days, m = 6. Find creep loss.

$$\theta = 1.6 \text{ (28 days)} \rightarrow \text{Loss} = 1.6 \times 6 \times 7.5 = 72 \text{ N/mm}^2$$

If the same member is transferred at 7 days : 2.2 × 6 × 7.5 = 99 N/mm². (Earlier transfer = larger creep.)

Elastic, creep and shrinkage losses are all strain-based – they depend on E_s. That is why high initial stress (HTS) is needed.

5. Relaxation of Stress in Steel

Relaxation is the **decrease of stress with time in steel held at constant strain** (length). Prestressing tendons are held at constant length after anchoring – their stress gradually decreases.

- It is a **steel property** (like creep of steel), depending on initial stress, temperature, time and type of steel. It increases with initial stress and temperature.
- **Pure relaxation** (tested at constant length) is higher than **actual (reduced) relaxation** in structure because the steel length changes slightly due to creep / shrinkage; IS gives test values for 1000 h.
- Low-relaxation (stabilised) strands are used now: loss ≈ ¼ to ½ of the normal-relaxation steel.

IS 1343 relaxation losses (at 1000 hours)

Initial prestress	Relaxation loss (N/mm²)
$0.5\,f_{pk}$	0
$0.6\,f_{pk}$	35
$0.7\,f_{pk}$	70
$0.8\,f_{pk}$	90

(f_{pk} = characteristic tensile strength of the steel. Interpolate for intermediate values. Values are for normal-relaxation steel at 20 °C.)

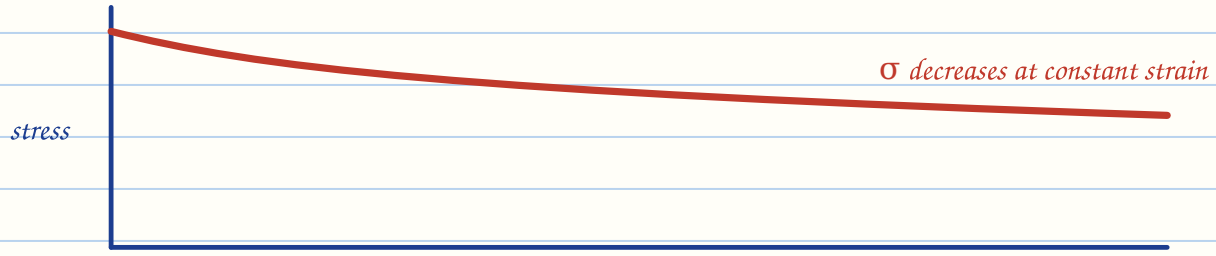


Fig. Relaxation of steel – stress vs time at constant length

Example 5

Initial stress in wires = 1200 N/mm², f_{pk} = 1600 N/mm². Find the relaxation loss.

$1200/1600 = 0.75\,f_{pk}$ → between 0.7 (70) and 0.8 (90)

Loss = 70 + (0.75 – 0.70)/(0.1) × 20 = **80 N/mm²** (6.7 % of 1200)

Relaxation loss ≈ 3 – 7 % of initial stress for normal steel; ≈ 1.5 – 2.5 % for low relaxation steel.

6. Anchorage Slip

Loss in **post-tensioned** members only. When the jack is released, the tendon draws a little into the anchorage as the wedges / cones seat themselves – a slip Δ (mm) – this reduces the tendon length (strain) → loss of stress.

- Typical slips (draw-in) : Freyssinet cone ≈ 4–6 mm; Gifford-Udall ≈ 1–2 mm; Magnel ≈ 1 mm; BBRV button head 0; threaded bars (Lee-McCall) 1 mm / nut.
- Slip influences a short length from the jacking end. Loss is greatest at the anchor and decreases towards the other end (friction also reverses).

Loss = (Δ / L) × E_s (for uniform strain; Δ in mm, L = length of tendon, mm)

- Δ = slip at anchorage (mm); L = length of the tendon (mm); E_s = modulus of steel.
- **Loss ∝ 1/L** – long tendons suffer less; short beams (small L) – larger %.
- Slip can be compensated by **over-tensioning** the jack by the extent of the expected slip.

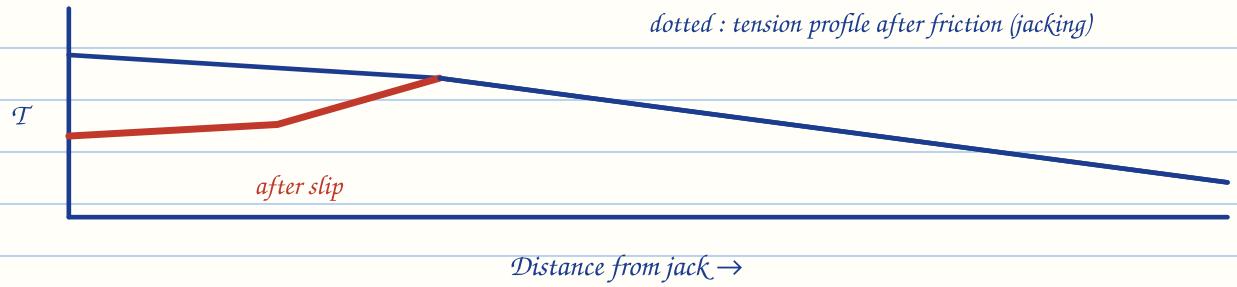


Fig. Effect of slip is confined to a length near the jack end

Example 6

A post-tensioned beam with tendons 20 m long is anchored with slip of 3 mm. $E_s = 2.1\times10^5$. Find loss of stress.

$$\text{Loss} = (3/20000) \times 2.1\times10^5 = 31.5\ N/mm^2$$

For L = 10 m – loss doubles (63 N/mm²). If initial stress = 1250 N/mm² → 2.5 % for 20 m.

7. Frictional Losses (post-tensioning only)

When tendon is stretched in a duct, friction between the tendon and duct reduces the force along its length. Force at the jack is largest, **smallest at the far (dead) end**.

Two components

1. **Length effect (wobble effect)** – because the duct can not be exactly straight; unintentional deviation. Depends on length x and wobble coefficient k.
2. **Curvature effect** – due to the intentional curvature of the duct; the tendon presses on the duct walls. Depends on the total angle turned α and coefficient of friction μ.

$$P_x = P_0 \cdot e^{-(\mu\alpha + kx)}$$

- P_0 = force at jack; P_x = force at distance x from jack; α = cumulative angle of curvature (radians) from jack to x;
- μ = coefficient of friction between tendon and duct; k = wobble coefficient per metre (also K).
- For small $(\mu\alpha + kx) \le 0.2$: $P_x \approx P_0 (1 - \mu\alpha - kx)$ (IS 1343 permits).

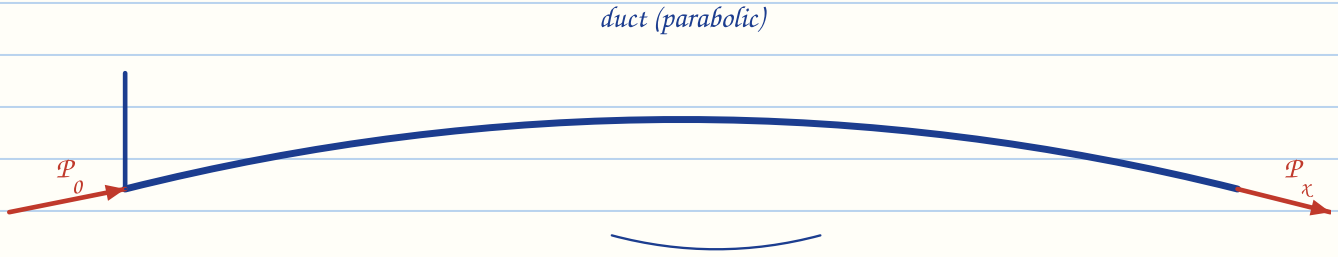


Fig. Tendon in a curved duct – friction reduces the force

IS 1343 coefficients

Type of duct / interface	μ
Steel moving on smooth concrete (formed with rubber tube)	0.55
Steel on steel (bright metal sheathing)	0.30
Steel on lead / greased	0.25

Wobble coefficient k = 0.0015 per metre (IS 1343, typical for sheathed ducts). Parabolic tendon : slope at end = $4e/L \rightarrow \alpha = 8e/L$ over whole span (for a symmetrical parabola with dip e).

Example 7 – Friction

A post-tensioned beam of span 20 m has a parabolic tendon with dip 0.4 m. P_0 = 1000 kN at the jack. μ = 0.35, k = 0.0015/m. Find the force and loss at the dead end.

$$\alpha = 8e/L = 8 \times 0.4/20 = 0.16 \text{ rad}$$

$$\mu\alpha + kx = 0.35 \times 0.16 + 0.0015 \times 20 = 0.056 + 0.030 = 0.086$$

$$P_x = 1000\, e^{-0.086} = 917.6 \text{ kN} \rightarrow \text{loss} = 82.4 \text{ kN } (8.24\ \%)$$

Approx. : 1000 (1 – 0.086) = 914 kN → loss 8.6 % (within accuracy).

Reducing friction : lubricate ducts, use bright metal sheathing, stress from both ends (loss at each end is half), use over-tensioning, mono-strands, use HDPE ducts, smaller curvature.

Stressing from both ends

Friction loss at the mid-span is lesser from both ends; the force profile is a "V" with equal values at both anchors. Thus the effective length for friction ≈ $L/2$ and loss halves. Commonly done for L > 25 m.

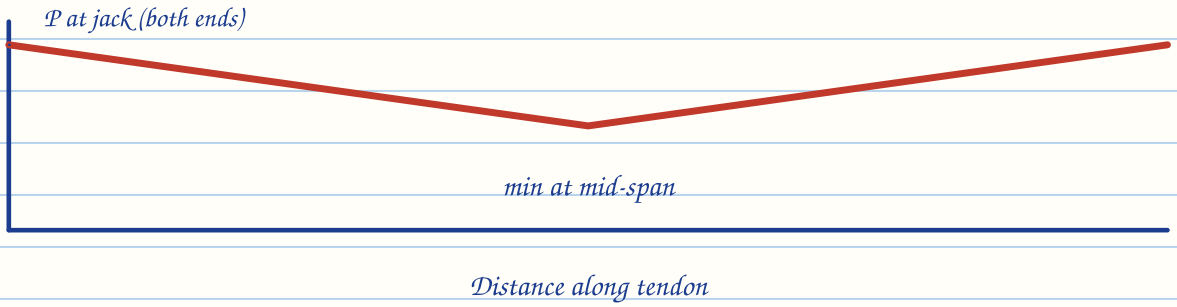


Fig. Tension profile when stressed from both ends

Frictional loss varies along length : 0 at jack, max at far end. Average loss ≈ half. Loss is automatically determined at design by the computed P_x at critical sections.

8. Total Losses – Summary

Type of loss	Pre-tensioned	Post-tensioned	Formula
Elastic shortening	Yes (full)	Partial (n cables)	$m f_c$; ½ $(n-1)/n\ m f_c$
Shrinkage	Yes (0.0003)	Yes (0.0002/log(t+2))	$\varepsilon_{cs} E_s$
Creep	Yes	Yes	$\theta\ m f_c$
Relaxation	Yes	Yes	IS table / % of f_{pi}
Anchorage slip	No	Yes	$(\Delta/L) E_s$
Friction	No	Yes	$P_0[1 - e^{-(\mu\alpha+kx)}]$

Typical percentage of total losses

Loss	Pre-tension	Post-tension
Elastic	3–5 %	1–3 %
Shrinkage	4–6 %	2–4 %
Creep	5–7 %	4–6 %
Relaxation	3–5 %	3–5 %
Slip	–	1–3 %
Friction	–	5–10 %
Total	**20–25 %**	**20–30 %**

IS 1343 (cl. 19.4) : If the losses are not calculated, a **lump-sum** may be assumed – generally **20 % for pre-tension** and **15 % for post-tension** (for concrete with relaxation low-level steel and no friction). This is only for preliminary design.

Effective prestress $P_e = P_i$ – (sum of losses) = $R \cdot P_i$

- **Time of loss :** elastic, friction, slip → at transfer (P_0). Creep, shrinkage, relaxation → over the years (P_e).
- Design checks : at transfer with P_0 (allowable stresses of concrete at transfer) and at service with $P_e = \eta\ P_0$.
- Typical **η = 0.80 – 0.85** (post-tensioned), 0.75 – 0.80 (pre-tensioned).

9. Worked Example – Pre-tensioned Beam (all losses)

A pre-tensioned beam 150 mm × 300 mm; 200 mm^2 of HTS wires at e = 50 mm. Initial stress = 1200 N/mm^2, f_{pk} = 1600 N/mm^2. f_{ck} = 40 N/mm^2, $E_s = 2.1 \times 10^5$. Transfer at 28 days. Estimate the total loss of prestress.

Step 1 – Section properties

$$A = 45000\ mm^2 ;\ I = 150 \times 300^3/12 = 3.375 \times 10^8\ mm^4$$

$$E_c = 5700\sqrt{40} = 36050\ N/mm^2 ;\ m = 2.1\times10^5/36050 = 5.83$$

$$P = 1200 \times 200 = 240\ kN$$

Step 2 – Stress in concrete at steel level

$$f_c = 240000/45000 + 240000 \times 50^2/3.375\times10^8 = 5.33 + 1.78 = 7.11\ N/mm^2$$

Step 3 – Losses

Loss	Working	N/mm²
Elastic shortening	$m f_c = 5.83 \times 7.11$	41.4
Shrinkage	$0.0003 \times 2.1\times10^5$	63.0
Creep (28 d, θ = 1.6)	$1.6 \times 5.83 \times 7.11$	66.3
Relaxation (0.75 f_{pk})	$70 + 0.5 \times 20$	80.0
Total		**250.7**

$$\%\ loss = 250.7/1200 \times 100 = 20.9\ \%$$

$$Effective\ prestress = 1200 - 250.7 = 949.3\ N/mm^2 \rightarrow P_e = 949.3 \times 200 = 189.9\ kN$$

Result matches the lump-sum 20 % rule for pre-tensioning. Residual factor η = 949.3/1200 = **0.79**.

10. Worked Example – Post-tensioned Beam

A post-tensioned beam 300 mm × 600 mm, span 20 m, parabolic tendon (ends 200 mm above the C.G., mid-span 200 mm below it, so dip = 0.40 m), A_{ps} = 800 mm², initial stress at jack 1250 N/mm², f_{pk} = 1770 N/mm². 4 cables stressed in sequence. f_{ck} = 50, $E_s = 2.1\times10^5$, μ = 0.35, k = 0.0015, slip = 3 mm, age at transfer = 28 d (shrinkage with t = 14 d). Find the total losses at the far end.

Section & material data

$$A = 1.8\times10^5 \text{ mm}^2;\ I = 300\times600^3/12 = 5.4\times10^9;\ E_c = 5700\sqrt{50} = 40305;\ m = 5.21$$

$$P = 1250\times800 = 1000 \text{ kN};\ f_c = 1000000/180000 + 10^6\times200^2/5.4\times10^9 = 5.56 + 7.41 = 12.96 \text{ N/mm}^2$$

Loss	Working	N/mm²
Elastic shortening	½ × (3/4) × 5.21 × 12.96	25.3
Shrinkage	$0.0002/\log_{10}(16)\times2.1\times10^5 = 1.661\times10^{-4}\times2.1\times10^5$	34.9
Creep (θ = 1.6)	1.6 × 5.21 × 12.96	108.1
Relaxation (0.706 f_{pk})	≈ 70 + 0.06 × 200 ≈ 70 (taken)	70.0
Anchorage slip	$3/20000\times2.1\times10^5$	31.5
Friction (α = 0.16, χ = 20)	$1250\,[1 - e^{-0.086}]$	103.0
Total		**372.8**

% loss = 372.8/1250 = 29.8 % (at far end). Effective stress = 877 N/mm²

Comments

- Friction and creep are the dominant losses in a long post-tensioned beam. The friction loss at the jack end is zero – the loss varies along the length; **design is done at critical section** (usually mid-span).
- At mid-span, friction ≈ 0.043 × 1250 ≈ 54 N/mm²; slip effect is zero if slip length < L/2 → total ≈ 290 N/mm² ≈ 23 %.
- Stressing from both ends would halve the friction loss and the slip effect of 31.5 N/mm² is retained at each end.

Always state clearly **where** (which section) the loss is being found. Note that creep depends on f_c at C.G. of steel.

11. Practice Problems & Important Questions

Numerical practice

1. A pre-tensioned beam 120 mm × 300 mm is prestressed by 4 wires Ø 7 mm at e = 60 mm, initial stress 1000 N/mm², m = 6. Find the loss due to elastic shortening. *(Ans ≈ 38 N/mm²)*
2. Estimate the shrinkage loss in (a) pre- (b) post-tensioned members of age 7 days for $E_s = 2\times10^5$. *(60 N/mm² ; 41.9 N/mm²)*
3. A concrete member transferred at 7 days has $f_c = 8$ N/mm² and m = 6. Find creep loss. *(2.2×6×8 = 105.6 N/mm²)*
4. A parabolic cable of 24 m span with dip 0.3 m and $P_0 = 800$ kN, $\mu = 0.3$, $k = 0.0015$. Find loss at the dead end. *($\alpha = 0.1; \Sigma = 0.066 \rightarrow 6.4$ %)*
5. A tendon of 15 m with slip 4 mm: find the loss. $E_s = 2\times10^5$. *(53.3 N/mm²)*

Short questions

1. What are the various types of losses in prestress ?
2. Why is there no friction loss in pre-tensioned members ?
3. Define relaxation, creep and shrinkage.
4. How is the loss due to elastic shortening reduced in post-tension ?
5. Mention the values of μ and k as per IS 1343.

Long questions

1. Explain in detail the various losses in pre-tensioned and post-tensioned members and the IS 1343 recommendations.
2. Derive the expression for friction loss $P_x = P_0 e^{-(\mu\alpha + kx)}$.
3. Compare the losses in pre- and post-tensioned members; mention how they can be minimised.
4. Numerical – compute all losses for a given beam (as in the worked example).

Formulae to memorise

Loss	Formula
Elastic (pre)	$m f_c$
Elastic (post)	$\frac{1}{2}\,(n-1)/n \cdot m f_c$
Shrinkage	$\varepsilon_{cs} E_s$; 0.0003 ; 0.0002/log(t+2)
Creep	$\theta\, m f_c$; θ = 2.2, 1.6, 1.1
Relaxation	IS 1343 table (0, 35, 70, 90)
Slip	$(\Delta/L) E_s$
Friction	$P_0\, e^{-(\mu\alpha + kx)}$

PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – III

FLEXURAL AND SHEAR

Syllabus : Analysis of beams for flexure and shear – beams pre-stressed with straight, concentric, eccentric, bent and parabolic tendons – kern line – cable profile – design of PSC beams (rectangular and I sections) using IS 1343 – analysis and design of rectangular and I beams for shear – introduction to transmission length and end block (no design and analytical problems).

Contents

1. Basic assumptions, notation, allowable stresses
2. Concentric tendon
3. Eccentric straight tendon
4. Bent & parabolic tendons – equivalent loads
5. Kern line & pressure line
6. Cable profile / cable zone
7. Design of rectangular beam (IS 1343)
8. Design of I-section beam
9. Shear in PSC beams – principal stresses, IS 1343
10. Shear design examples
11. Transmission length & end block (introduction)
12. Important questions & formula sheet

Exam tip : Design problems carry 10–15 marks. Follow the steps : Z_b → select section → P and e → check stresses at transfer and service.

1. Analysis – Basic Assumptions

1. Concrete is a **homogeneous, elastic** material – plane sections remain plane (Bernoulli).
2. The section is **un-cracked** under working loads (Type 1 / 2) – gross section properties A, I, Z used (or transformed).
3. Steel (tendon) force P is applied at the tendon centroid at eccentricity e from the C.G. of concrete section.
4. Stresses are found by **superposition** of (i) direct stress P/A, (ii) bending due to prestress $P{\cdot}e$, (iii) bending due to loads M.
5. Losses are accounted by $\eta = P_e/P_i$ (0.75 – 0.85).

Notation

Symbol	Meaning
P_i, P_0, P_e	initial, at transfer, effective prestress force
A, I	area & M.I. of the section
y_t, y_b	distance of top / bottom fibre from C.G.
Z_t, Z_b	section moduli I/y_t, I/y_b
e	eccentricity of tendon below C.G.
M_g, M_q, M_T	moments : self weight, imposed, total
r	radius of gyration, $r^2 = I/A$

General stress equation (compression +ve)

$$f = P/A \pm P\,e\,y/I \pm M\,y/I$$

$$\text{Top : } f_t = P/A - Pe/Z_t + M/Z_t \quad \text{Bottom : } f_b = P/A + Pe/Z_b - M/Z_b$$

Permissible stresses (IS 1343)

- At **transfer** : compression ≤ 0.54 f_{ci} (flexure); tension = 0 for Type 1 (small tension permitted in Type 2 as per code table).
- At **service** : compression ≤ 0.41 f_{ck} (flexure); tension = 0 (Type 1), limited tension (Type 2 – see IS 1343 Table 5).
- f_{ci} = cube strength at transfer. Minimum grade : M40 pre-tension, M30 post-tension.
- In design problems the permissible values are usually **given**; use them as given.

2. Beam with Concentric (Axial) Tendon

The tendon passes through the C.G. of the section: e = 0. Prestress produces a **uniform compression** P/A – no camber.

$$f = P/A \pm M y / I$$

Example 1

A rectangular beam 200 mm × 400 mm, span 6 m, carries total UDL 12 kN/m (including self-weight). Concentric P = 600 kN. Find extreme stresses at mid-span.

$$A = 80000\ mm^2 ;\ Z = 200 \times 400^2/6 = 5.333 \times 10^6\ mm^3 ;\ M = 12 \times 6^2/8 = 54\ kN{\cdot}m$$

$$P/A = 7.5\ N/mm^2 ;\ M/Z = 54\times10^6 / 5.333\times10^6 = 10.125\ N/mm^2$$

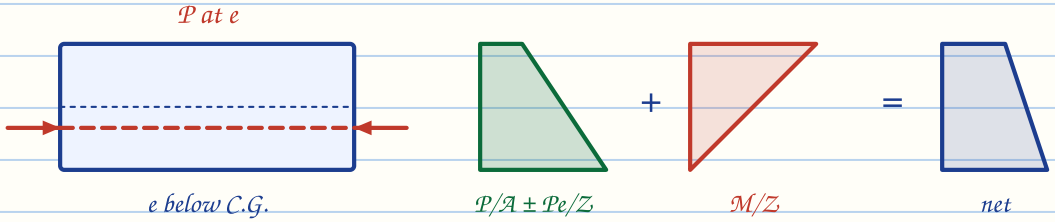
$$Top = 7.5 + 10.125 = 17.63\ N/mm^2\ (comp.) ;\ Bottom = 7.5 - 10.125 = -2.63\ N/mm^2\ (tension)$$

Net tension at bottom → cracking. Hence give an **eccentricity** – next page. Concentric prestress is used in axially loaded members (piles, poles, pipes, tension ties).

3. Beam with Eccentric Straight Tendon

Tendon is at a constant eccentricity e **below** the C.G. – the prestress now has a hogging moment P·e which **counteracts the sagging load moment**.

$$f_t = P/A - P\,e/Z_t + M/Z_t \qquad f_b = P/A + P\,e/Z_b - M/Z_b$$



Top fibre : compression from M, tension from Pe → balanced; bottom : Pe compression – M tension

Stress is generally compression throughout when e is chosen well

Fig. Eccentric prestress – stress diagrams

Example 2 (same beam, e = 60 mm)

$$P\,e/Z = 600\times10^3 \times 60 / 5.333\times10^6 = 6.75\ N/mm^2$$

$$Top = 7.5 - 6.75 + 10.125 = 10.88\ N/mm^2 ;\ Bottom = 7.5 + 6.75 - 10.125 = 4.13\ N/mm^2$$

Both fibres are in compression – no tension. **Check at transfer (self-weight only, 2 kN/m → M_g = 9 kN·m, M_g/Z = 1.69)** : top = 7.5 – 6.75 + 1.69 = 2.44 ✓ (no tension).

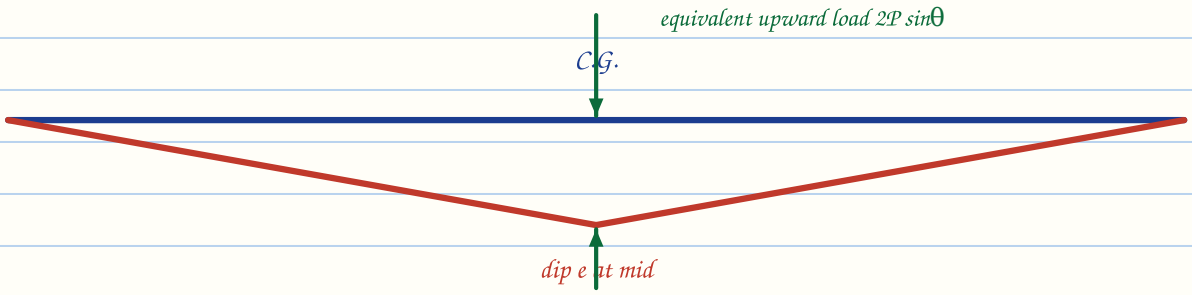
At the ends (M = 0) an eccentric straight tendon gives top tension $P\,e/Z_t - P/A$ (here 6.75 – 7.5 = –0.75 → still compressive) – if e is large, tension appears at the **top of the end sections**; remedy : (i) reduce e near the supports (bent/harped or parabolic tendon), (ii) de-bond some strands, (iii) provide top steel.

Eccentric straight tendon : simple, used in pre-tensioned beams, slabs. For longer spans use harped / parabolic tendons.

4. Bent (Harped) and Parabolic Tendons

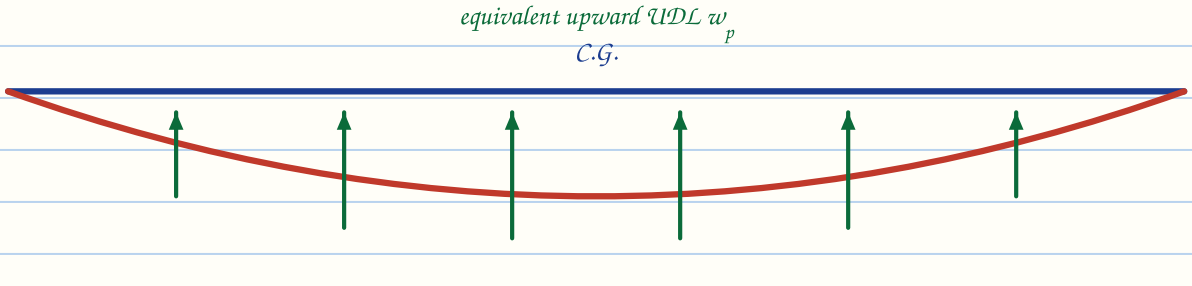
Tendon profile is varied so that e is large at mid-span (max moment) and small (zero) at supports – lesser end stresses, **vertical component of prestress reduces shear**.

(a) Bent (harped) tendon – kink at mid-span or third points



- Angle of the tendon $\theta \approx \tan\theta = 2e/L$ (end at C.G., e at midspan).
- Equivalent **upward concentrated load at the kink** $W = 2P \sin\theta \approx 2P(2e/L) = \mathbf{4Pe/L}$.
- Moment at mid-span from prestress = P·e (sagging comp. – hogging); **vertical shear component** $V_p = P \sin\theta$.

(b) Parabolic tendon – equivalent UDL



$$y = 4ex(L-x)/L^2;\ d^2y/dx^2 = -8e/L^2 \rightarrow w_p = 8Pe/L^2 \text{ (upward)}$$

- Total upward force = $w_p L = 8Pe/L$ = twice the end slope force : slope at end $4e/L \rightarrow$ vertical component $4Pe/L$ each end ✓.
- Net moment from prestress at section = P·y (y = tendon dip at that section).
- Net downward load = $w - w_p \rightarrow$ if $w = w_p$ the beam is perfectly balanced (load balancing).

5. Equivalent Loads – Summary

Tendon profile	Equivalent load	Moment due to P
Straight, concentric	No transverse load; axial P only	$M = 0$
Straight, eccentric e	End moments P·e (hogging)	$M = -P\,e$ (constant)
Bent (single kink, dip e)	Upward point load $4Pe/L$ at mid; end axial P	$M = -P\cdot y(x)$
Parabolic, dip e	Upward UDL $8Pe/L^2$; end reactions	$M = -P\cdot y(x)$
Circular arc	Radial load P/R per unit length	$M = -P\cdot y$

Example 3 (parabolic tendon, load balancing)

Beam of Example 1 (6 m, 200 × 400) with a parabolic tendon, e = 0 at ends and e = 100 mm at mid, P = 600 kN, total load 12 kN/m. Find stresses at mid-span and the net load.

$$w_p = 8 \times 600 \times 0.1/6^2 = 13.33 \text{ kN/m (upward)} > 12 \text{ kN/m} \rightarrow \text{net upward } 1.33 \text{ kN/m}$$

$$\text{Top} = 7.5 - 600000\times100/5.333\times10^6 + 10.125 = 7.5 - 11.25 + 10.125 = 6.38 \text{ N/mm}^2$$

$$\text{Bottom} = 7.5 + 11.25 - 10.125 = 8.63 \text{ N/mm}^2$$

Both compressive. At the support $M = 0$, $e = 0 \rightarrow$ uniform $P/A = 7.5$ N/mm² ✓ (no end tension).

Cambering : at transfer, only self-weight (say 2.4 kN/m) acts but the full upward 13.33 kN/m is present → large camber. Check top fibre tension at transfer: $M_g = 10.8$ kN·m → top = 7.5 − 11.25 + 2.03 = −1.72 N/mm² (tension) → provision of top steel or lesser e needed.

Check **two stages** always : (1) transfer – P_0 with M_g only; (2) service – η P_0 with M_T.

6. Kern Line and Pressure Line

Kern of a section

The **kern** is the region within which a compressive load (prestress) can be applied **without producing tension** anywhere in the section.

- For a load P at eccentricity e : $f = P/A\ (1 \pm e\,y/r^2)$. Tension is absent if $e\,y / r^2 \le 1$.
- Upper kern distance (above C.G.) : $k_t = Z_b / A = r^2 / y_b$.
- Lower kern distance (below C.G.) : $k_b = Z_t / A = r^2 / y_t$.

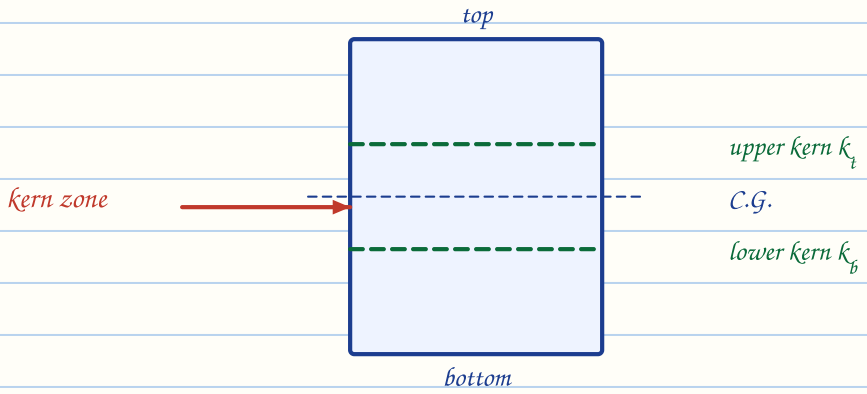


Fig. Kern zone (middle-third for rectangle)

Rectangle $b \times D$: $k_t = k_b = D/6$; I-section : $k_t = Z_b/A$, $k_b = Z_t/A$

Pressure line (thrust line, C-line)

- The locus of the **resultant compressive force** in the concrete (C-line). It is the line of action of the compressive force in concrete – for a concordant tendon C-line coincides with the cable.
- Distance a from the cable (C-line shift) : $a = M / P$ (due to external moments, C-line moves up by M/P).
- **For no tension**, the C-line must stay within the kern at all sections.

C-line shift due to load : $a = M / P$; C-line position = e – M/P (above cable)

Example 4 – Kern

Rectangular section 200 × 400. Find the kern points and check that e = 60 mm lies in the kern.

$$k = D/6 = 400/6 = 66.7 \text{ mm above and below the C.G.}$$

$$e = 60 \text{ mm} < 66.7 \text{ mm} \rightarrow \text{inside kern} \rightarrow \text{no tension from } P \text{ alone} \checkmark$$

Check : stress at the top fibre = $P/A\,(1 - 6e/D) = 7.5\,(1 - 0.9) = +0.75\ N/mm^2$ ✓ compression.

7. Cable Profile and Cable Zone

Cable zone is the zone in the depth of the beam within which the tendon profile (centroid) can be placed such that stresses are within permissible limits at all sections. Beyond it, tension / over-compression results.

Limiting eccentricities – at a section with M_g (transfer) and M_T (service)

- **Transfer** : (i) bottom fibre compression ≤ f_{ci}; (ii) top fibre tension ≥ f_{ti} (0 for Type 1).
- **Service** : (iii) top fibre compression ≤ f_{cw}; (iv) bottom fibre tension ≥ f_{tw} (0 for Type 1).

$$(1)\ e \le Z_b(f_{ct} - P_i/A)/P_i + M_g/P_i$$

$$(2)\ e \le Z_t(P_i/A - f_{ti})/P_i + M_g/P_i$$

$$(3)\ e \ge M_T/(\eta P_i) - Z_t(f_{cw} - \eta P_i/A)/(\eta P_i)$$

$$(4)\ e \ge M_T/(\eta P_i) + Z_b(f_{tw} - \eta P_i/A)/(\eta P_i)$$

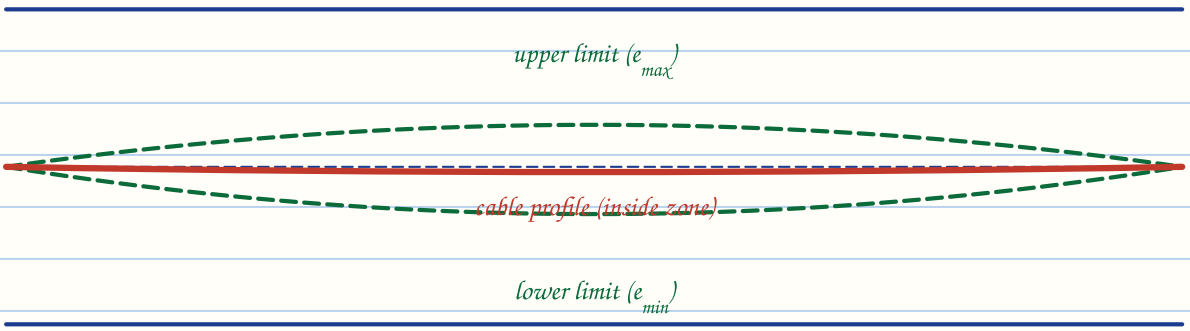


Fig. Cable zone for a simply supported beam

Concordant cable : cable profile such that the tendon stays at the C-line (no secondary moments). In a simply supported beam **any cable is concordant**; in continuous beams only some profiles are (secondary moments arise otherwise).

8. Design of PSC Beam – Rectangular Section (IS 1343)

Procedure (Type 1 – no tension, simply supported beam)

1. Assume section depth $D \approx L/20 \ldots L/25$, estimate self-weight → M_g, then M_q and $M_T = M_g + M_q$.
2. Required **section modulus** :

$$Z_b \geq (M_T - \eta M_g)/(\eta f_{ct} - f_{tw})$$

- f_{ct} = permissible compression at the bottom at transfer ; f_{tw} = permissible tension at the bottom at service (0 for Type 1).
- Derivation : eliminate $P/A + Pe/Z_b$ from the bottom fibre equations at transfer and at service.
- Similarly for the top fibre : $Z_t \geq (M_T - \eta M_g)/(f_{cw} - \eta f_{tt})$.

1. Choose the section (b, D) with $Z \geq$ required Z; recompute M_g with the actual self-weight.
2. Find P_i and **e** : $P_i (1/A + e/Z_b) = M_T/(\eta Z_b)$ (for $f_{tw} = 0$) — choose e (practical : y_b – cover) to get P_i.
3. Check stresses at **transfer** and **service** at mid-span (top & bottom) and at the **support**.
4. Choose steel : $A_{ps} = P_i/f_{pi}$ ($f_{pi} \approx 0.6$–$0.7 f_{pk}$).
5. Check ultimate moment of resistance $M_u \geq 1.5 M_T$ (IS 1343 – load factor), shear, deflection, end block, etc.

Derivation of Z_b : Transfer : $P_i/A + P_i e/Z_b - M_g/Z_b = f_{ct}$. Service : $\eta(P_i/A + P_i e/Z_b) - M_T/Z_b = f_{tw}$. Substituting → $Z_b = (M_T - \eta M_g)/(\eta f_{ct} - f_{tw})$.

Design Example 5 – Rectangular beam

Design a simply supported post-tensioned beam of span 10 m for an imposed load of 10 kN/m. Type 1 (no tension) ; f_{ct} = 15 N/mm² at transfer (bottom), f_{cw} = 16 N/mm² at service (top), f_{tt} = f_{tw} = 0 ; η = 0.8 ; concrete M40, density 25 kN/m³.

Step 1 – Trial section 300 × 500

$$A = 1.5\times10^{5}\ mm^2\ ;\ Z = 300\times500^2/6 = 12.5\times10^{6}\ mm^3\ ;\ self\ weight = 3.75\ kN/m$$

$$M_g = 3.75\times10^2/8 = 46.9\ kN\cdot m\ ;\ M_q = 10\times10^2/8 = 125\ kN\cdot m\ ;\ M_T = 171.9\ kN\cdot m$$

$$Z_{req} = (171.9 - 0.8\times46.9)\times10^6/(0.8\times15 - 0) = 134.4\times10^6/12 = 11.2\times10^6 < 12.5\times10^6\ \checkmark\ OK$$

Step 2 – Prestressing force and eccentricity

$$P_i(1/A + e/Z) = M_T/(\eta Z) = 171.9/(0.8\times12.5) = 17.19\ N/mm^2$$

Adopt e = 120 mm (cover and duct limit; larger e gives top tension at transfer) :

$$1/A + e/Z = 6.667\times10^{-6} + 9.6\times10^{-6} = 1.627\times10^{-5} \rightarrow P_i = 17.19/1.627\times10^{-5} = 1057\ kN$$

Step 3 – Check stresses at mid-span (N/mm²)

Stage	Top fibre	Bottom fibre
Transfer (P_i, M_g)	7.05 − 10.15 + 3.75 = **+0.65** ✓	7.05 + 10.15 − 3.75 = **13.45** ≤ 15 ✓
Service (ηP_i, M_T)	0.8(7.05 − 10.15) + 13.75 = **11.27** ≤ 16 ✓	0.8(17.20) − 13.75 = **≈ 0** ✓

Step 4 – Steel

Initial stress ≈ 1070 N/mm² (0.6 f_{pk} for f_{pk} = 1770) → A_{ps} = 1057×10³/1070 = 988 mm² → **10 strands Ø 12.7 mm (987 mm²)**. Ducts : 1 cable of 10 strands (or 2 × 5).

Step 5 – Support section

At the support M = 0 → tendon should be at C.G. (e = 0) or within kern (e ≤ D/6 = 83 mm) : uniform P/A = 7.05 N/mm² ✓ . Use a **parabolic profile** from e = 120 mm at mid to e = 0 at supports (or raised cable).

> Final : 300 × 500 mm beam, P_i = 1057 kN, e = 120 mm at mid-span, 10 strands Ø 12.7 mm, stirrups by shear design (Unit III shear).

9. Design of I-Section Beam

For longer spans an I-section (symmetrical / unsymmetrical) is more efficient: the **flanges** carry the stresses, the thin web is used mainly for shear – high Z/A ratio → larger kern, smaller P.

Efficiency of the section : $k_t + k_b = (Z_t + Z_b)/A$ (larger = better)

Design Example 6

Design an I-section beam, simply supported, span 14 m, imposed load 12 kN/m. Trial section : flanges 400 × 100 each, web 100 × 600, D = 800 mm. Permissible : $f_{ct} = 15$, $f_{cw} = 16$, $f_{tw} = f_{tt} = 0$; $\eta = 0.8$.

$$A = 2\times400\times100 + 100\times600 = 1.4\times10^5 \text{ mm}^2$$

$$I = 400\times800^3/12 - 300\times600^3/12 = 1.7067\times10^{10} - 5.4\times10^9 = 1.1667\times10^{10} \text{ mm}^4$$

$$Z_t = Z_b = I/400 = 29.17\times10^6 \text{ mm}^3 \text{ ; self weight} = 0.14 \times 25 = 3.5 \text{ kN/m}$$

$$M_g = 3.5 \times 14^2/8 = 85.75 \text{ kN·m} ; M_q = 12 \times 14^2/8 = 294 \text{ kN·m} ; M_T = 379.75 \text{ kN·m}$$

$$Z_{req} = (379.75 - 0.8\times85.75)/(0.8\times15) = 311.15/12 = 25.9\times10^6 < 29.17\times10^6 \checkmark$$

Prestressing force & eccentricity

$$P_i(1/A + e/Z) = M_T/(\eta Z) = 379.75/(0.8 \times 29.17) = 16.27 \text{ N/mm}^2$$

Max practical $e = y_b - \text{cover}(150) = 400 - 150 = 250$ mm : $1/A + e/Z = 7.143\times10^{-6} + 8.57\times10^{-6} = 1.571\times10^{-5}$

$$P_i = 16.27/1.571\times10^{-5} = 1036 \text{ kN}$$

Stage	Top fibre (N/mm²)	Bottom fibre (N/mm²)
Transfer	7.40 − 8.88 + 2.94 = **+1.46** ✓	7.40 + 8.88 − 2.94 = **13.34** ≤ 15 ✓
Service	0.8(7.40−8.88) + 13.02 = **11.84** ≤ 16 ✓	0.8(16.28) − 13.02 = **≈ 0** ✓

Steel : $A_{ps} = 1036\times10^3/1100 = 942$ mm² → **10 strands Ø 12.7** (987 mm²). Profile : parabolic, e = 250 mm at mid-span reducing to 0 – 100 mm at supports (within kern $k = Z/A = 208$ mm).

10. Analysis of Shear in PSC Beams

Shear stress in PSC beams must be checked for **principal tension**, which causes diagonal cracking. Prestress (axial compression) reduces the principal tension.

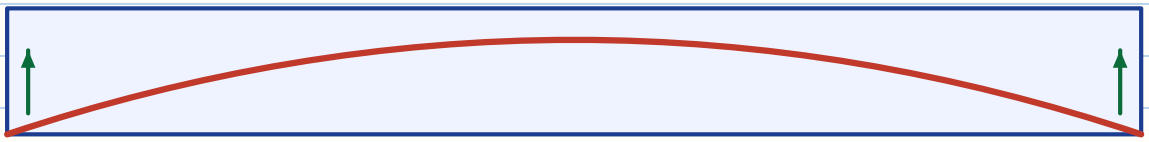
Principal stresses

$$\sigma_{1,2} = f/2 \pm \sqrt{[(f/2)^2 + \tau^2]} \quad \tan 2\alpha = 2\tau / f$$

- f = longitudinal compressive stress ($P/A \pm Pe\,y/I \pm M\,y/I$) ; τ = shear stress = $V\,S/(I\,b)$ (neglecting vertical stress).
- Principal tension $\sigma_t = -\sigma_2 = \sqrt{[(f/2)^2 + \tau^2]} - f/2$. Without prestress ($f = 0$) $\sigma_t = \tau \rightarrow$ with prestress it is less.
- Cracking when σ_t reaches tensile strength $f_t = 0.24\sqrt{f_{ck}}$ (direct tension).
- Maximum principal tension is normally at the **junction of web and flange** or at the C.G. of I-section.

Effect of tendon curvature

- Inclined (harped / parabolic) tendon gives a vertical component $V_p = P \sin\theta$ opposing the external shear.
- **Net shear** $V_{net} = V - V_p$; shear stress $\tau = (V - P \sin\theta)\,S/(I\,b)$.



$V_p = P \sin\theta$ opposes V

Fig. Tendon curvature reduces net shear

Types of cracks : (i) **flexure-shear cracks** – flexural crack propagates diagonally (cracked section, V_{cr}); (ii) **web-shear cracks** – principal tension in web exceeds f_t before flexure cracks (uncracked, V_{co}), common near supports and in thin webs.

IS 1343 gives shear resistance in two cases : V_{co} (un-cracked in flexure) and V_{cr} (cracked in flexure). The **lesser** controls.

11. Shear Design as per IS 1343

(a) Section uncracked in flexure (V_{co})

$$V_{co} = 0.67\, b\, D \sqrt{(f_t^2 + 0.8\, f_{cp} f_t)} \; (+ V_p)$$

- $f_t = 0.24 \sqrt{f_{ck}}$ (max principal tensile stress) ; f_{cp} = compressive stress at centroidal axis due to prestress (P_e/A) ; b = width of web ; D = overall depth.
- Check where there is no flexural tension in the section at ultimate (e.g., near supports).

(b) Section cracked in flexure (V_{cr})

$$V_{cr} = 0.037\, b\, d \sqrt{f_{ck}} + M_0\, V / M \geq 0.1\, b\, d \sqrt{f_{ck}}$$

- M_0 = moment necessary to produce zero stress at the extreme tension fibre = $0.8\, f_{pt}\, I / y_t$; f_{pt} = stress at tension fibre due to prestress only ; V, M = shear and moment at ultimate at the section ; d = effective depth.
- V_{cr} is calculated at sections more than $D/2$ from the support, and V_c = lesser of V_{co} and V_{cr} (where both apply).

Shear reinforcement

- If $V \leq V_c$ provide only **minimum (nominal) shear reinforcement**; if $V > V_c$ design stirrups for $(V - V_c)$.
- Vertical stirrups :

$$A_{sv} / s_v = (V - V_c) / (0.87\, f_y\, d_t)$$

- Minimum : $A_{sv} / (b\, s_v) \geq 0.4 / (0.87\, f_y)$ (IS 1343 cl. 22.4.2).
- Max spacing $s_v \leq 0.75\, d_t$ (or 4 b) and ≤ 450 mm.
- d_t = depth to the centroid of the stirrup – i.e. distance from extreme compression fibre to the longitudinal tension steel.
- Shear stress is limited to $\tau_{c,max}$ (IS 1343 Table 6) – otherwise enlarge the web.

Example 7 – Shear (uncracked)

The beam of Design Example 5 (300 × 500, M40, $P_e = 0.8 \times 1057 = 845$ kN, tendon at C.G. at the supports). Total service load = 13.75 kN/m, span 10 m. Check shear at the support section.

$$V_u = 1.5 \times 13.75 \times 10/2 = 103 \text{ kN}; \; f_{cp} = 845\times10^3/1.5\times10^5 = 5.63 \text{ N/mm}^2$$

$$f_t = 0.24\sqrt{40} = 1.52 \text{ N/mm}^2$$

$$V_{co} = 0.67 \times 300 \times 500 \times \sqrt{(1.52^2 + 0.8\times5.63\times1.52)} = 100500 \times \sqrt{(2.31 + 6.85)} = 100500 \times 3.03$$

$$= 304 \text{ kN}$$

$V_u = 103$ kN < $V_{co} = 304$ kN → only **minimum shear reinforcement** is needed : $A_{sv}/(b\, s_v) \geq 0.4/(0.87 \times 415) = 0.00111$. 2-legged 8 mm stirrups (101 mm²): $s_v = 101/(0.00111 \times 300) = 303$ mm → limited to 0.75 $d_t \approx 280$ → **provide 8 mm @ 250 mm c/c**.

Example 8 – Shear (stirrups required)

A beam web $b = 300$, $D = 500$, $d_t = 400$, $V_c = 304$ kN (above), ultimate shear $V = 450$ kN at a section. $f_y = 415$ N/mm². Design stirrups.

$$A_{sv}/s_v = (450 - 304)\times10^3 / (0.87 \times 415 \times 400) = 146000/144420 = 1.01 \text{ mm}^2/\text{mm}$$

2-legged 10 mm : $A_{sv} = 157$ mm² → $s_v = 157/1.01 = 155$ mm → **provide 10 mm @ 150 mm c/c**

Check $s_v \leq 0.75 \times 400 = 300$ mm ✓ ; minimum $A_{sv}/s_v = 0.00111 \times 300 = 0.33$ mm²/mm ✓ (1.01 > 0.33).

Example 9 – V_{cr} (cracked section)

At a section : $M = 150$ kN·m, $V = 120$ kN (ultimate), $b = 300$, $d = 370$, $f_{ck} = 40$, $f_{pt} = 8$ N/mm² (prestress tension fibre = bottom), $I = 3.125 \times 10^9$, $y_t = 250$.

$$M_0 = 0.8 \times 8 \times 3.125\times10^9/250 = 80 \text{ kN·m}$$

$$V_{cr} = 0.037 \times 300 \times 370 \times \sqrt{40} + 80 \times 120/150 = 26.0 + 64.0 = 90 \text{ kN}$$

Minimum : $0.1 \times 300 \times 370 \times 6.32 = 70.2$ kN ✓ . Since $V = 120 > V_{cr} = 90$ kN, stirrups for 30 kN : $A_{sv}/s_v = 30000/(0.87\times415\times370) = 0.22$ → provide minimum (0.33).

> Take V_c as the **lesser** of V_{co} and V_{cr} and always provide minimum stirrups.

12. Transmission Length and End Block (Introduction)

Transmission length (pre-tensioned members)

- In pre-tensioned members the stress is transferred from tendon to concrete **by bond**. At the free end the steel stress is zero; it builds up to full value over a length called the **transmission length L_t**.
- Within L_t the prestress is less than the effective value → the shear & flexure design near supports must consider reduced prestress.
- Factors affecting L_t : diameter & type of tendon (strands, indented wires), concrete strength at transfer, surface condition, rate of release (gradual – sudden increases L_t), Hoyer effect, tendon stress.
- IS 1343 : $L_t \approx 30\,\varphi - 100\,\varphi$ depending on the type; some practical values: plain wires ≈ 100 φ ; strands ≈ 30–60 φ; (φ = diameter).

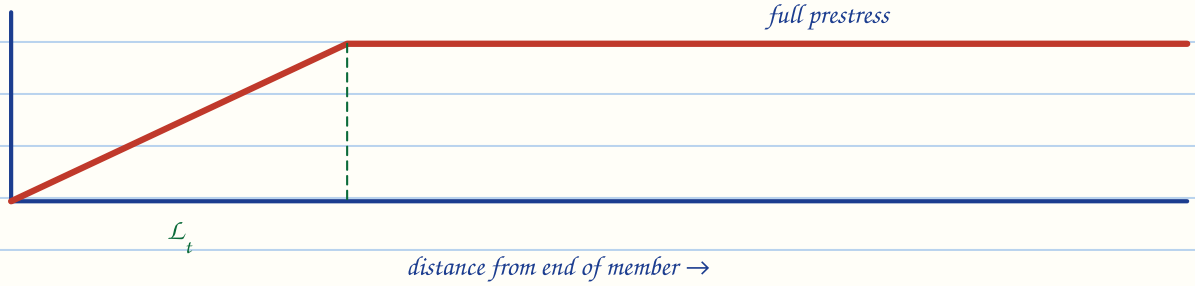


Fig. Variation of steel stress over the transmission length

End block (post-tensioned members)

- At the end of a post-tensioned member the anchorage force P is applied as a **concentrated load** on a small area. The stress spreads over the full section only after a distance ≈ depth of the member – this length is the **end block** (anchorage zone).
- Within the end block high **transverse tensile stresses** arise : **bursting tension** (centre of block, along load axis) and **spalling tension** (at the loaded face corners/edges); also high bearing stresses under the plate.
- Reinforcement : closely spaced **helices / stirrups / mesh** (bursting steel) designed by Guyon / Magnel methods; bearing plates, local enlargement of the end. (Design not required in this syllabus.)

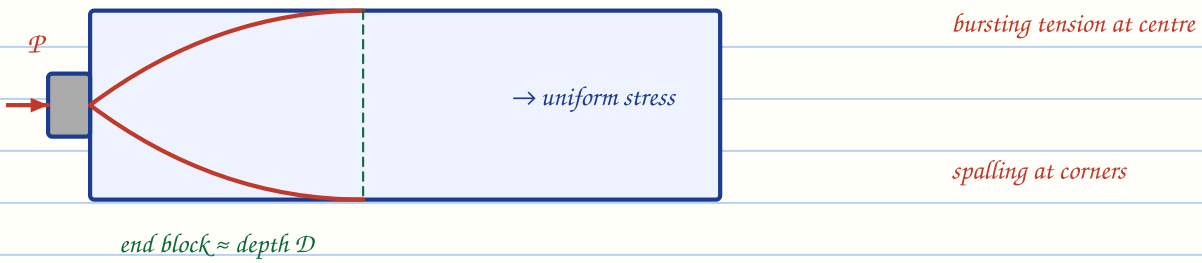


Fig. Anchorage zone (end block) – spread of prestress

13. Important Questions & Formula Sheet

Short answers

1. State the assumptions in the analysis of PSC section.
2. What is the kern ? Locate kern points for a rectangular section.
3. What is cable zone ? What is a concordant cable ?
4. Write the equivalent load for a parabolic tendon.
5. What is transmission length ? Define end block.
6. Why does prestress reduce principal tension ?

Long answers

1. A PSC beam 200×400 with a parabolic tendon … find stresses at mid-span and at supports (analysis).
2. Design a PSC rectangular beam for given span and load using IS 1343 (Z_b procedure).
3. Design an I-section PSC beam – check stress at transfer and service.
4. Explain the shear failure of PSC beams. Derive V_{co} and V_{cr} (IS 1343) and design shear reinforcement.
5. Explain the concept of load balancing; write about kern points and thrust line.

Formulae

Item	Formula
Stress	$f = P/A \pm Pe\,y/I \pm M\,y/I$
Kern	$k_t = Z_b/A$; $k_b = Z_t/A$ ($D/6$ for rect.)
Parabolic equivalent UDL	$w_p = 8Pe/L^2$
Bent tendon	$W = 4Pe/L$; $V_p = P \sin\theta$
Design Z_b	$(M_T - \eta M_q)/(\eta f_{ct} - f_{tw})$
V_{co}	$0.67\,b\,D\sqrt{(f_t^2 + 0.8\,f_{cp}\,f_t)}$
V_{cr}	$0.037\,b\,d\sqrt{f_{ck}} + M_0 V/M$
Stirrups	$A_{sv}/s_v = (V - V_c)/(0.87\,f_y\,d_t)$

Note : Permissible stress values and shear coefficients should be taken from IS 1343 (Tables 5 & 6, cl. 22) as given in your question paper / data book.

PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – IV
DEFLECTIONS

Syllabus : Control of deflections – factors influencing deflections – short-term deflections of un-cracked beams – prediction of long-time deflections.

Contents

1. Introduction – need to control deflection; IS limits
2. Factors influencing deflections
3. Deflection of beams – basic formulae
4. Deflection due to prestress – straight, bent, parabolic tendons
5. Worked examples (short-term)
6. Cracked sections – brief note
7. Long-term deflections – creep, shrinkage, loss of prestress
8. Worked example – long-term deflection
9. Control of deflections – measures
10. Important questions & formula sheet

Exam tip : Learn the deflection formula for each tendon profile ($Pe\,L^2/8EI$, $5PeL^2/48EI$, $PeL^2/12EI$) – every numerical uses them.

1. Introduction

PSC members are slender and made of high-strength materials – so **serviceability (deflection)** often governs the design rather than strength.

- Prestress produces an **upward deflection (camber)** that opposes the downward deflection due to loads; the net deflection may be upward or downward.
- Excess camber or deflection leads to: cracking of partitions / finishes, ponding on roofs, drainage problems, vibration, machine misalignment, poor appearance and loss of user confidence.
- Deflection is a **serviceability limit state** – checked under working loads (and for long-term effects).

IS 1343 / IS 456 limits (cl. 23.2)

Case	Limit
Final deflection (due to all loads, incl. creep, shrinkage, temperature)	≤ span / 250
Deflection after erection of partitions & finishes (incl. creep, shrinkage, temperature)	≤ span / 350 or 20 mm, whichever is less
Cantilevers	span replaced by 2 × projection
Camber (upward) – practical	excess camber to be avoided (≈ span/300 – span/250 at transfer)

Camber / deflection in a PSC beam

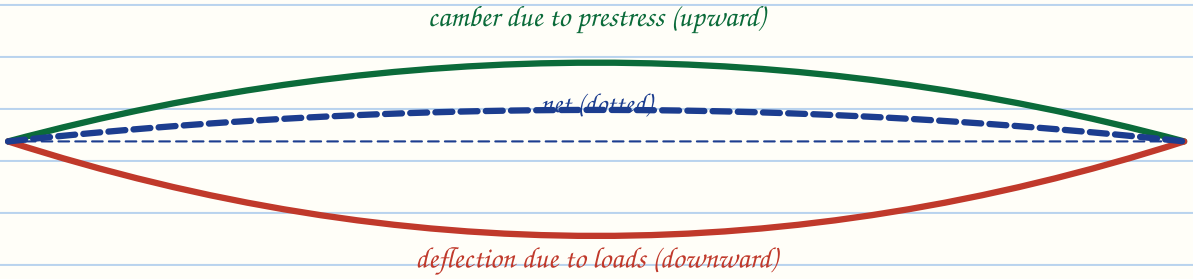


Fig. Upward camber from prestress opposes downward load deflection

Net deflection = (deflection due to loads) – (camber due to prestress). At transfer only self-weight acts while P is maximum → **maximum camber** occurs at transfer.

2. Factors Influencing Deflections

Factor	Effect on deflection
Magnitude of prestressing force P	$\uparrow P \rightarrow \uparrow$ camber (upward deflection)
Eccentricity & profile of tendon	Large e at mid-span, parabolic/bent profile → more camber; shape decides the coefficient (1/8, 5/48, 1/12)
Load : self weight, dead, live; their type and duration	Deflection ∝ load; sustained loads increase with creep
Span L	Short-term $\delta \propto L^2$ (prestress) or L^4 (UDL) – increases rapidly with span
Flexural rigidity $E_c I$	Higher grade concrete → $\uparrow E_c$; deeper section → $\uparrow I$; cracked section → lower I
Modulus of elasticity of concrete E_c (age)	$E_c = 5700\sqrt{f_{ck}}$ (IS 1343) ; E_c at transfer is smaller (f_{ci}) → larger camber at transfer
Creep of concrete	Deflection increases with time (factor $1+\theta$) ; camber also grows with time
Shrinkage of concrete	Curvature (unsymmetric steel) → additional deflection
Loss of prestress (creep, shrinkage, relaxation)	Reduces camber with time → net deflection increases
Cracking of concrete	Reduces effective I → deflection suddenly increases after M_{cr}
Support conditions & continuity	Fixed ends / continuity reduce deflection ($M_{support}$ hogging)
Temperature & humidity, age at loading	Affect creep / shrinkage → long-term deflection
Type of prestressing / bond	Un-bonded vs bonded – changes stress in steel at overload

Summary : δ depends on **(P, e, profile) × (L^2/EI)** and the **load history** through creep. Controlling P, e and $E_c I$ controls δ.

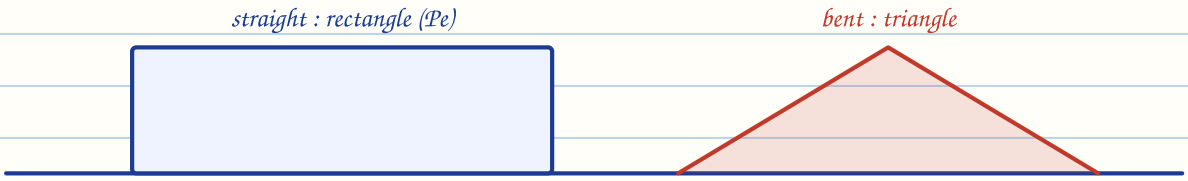
3. Deflection of Beams – Basic Formulae

All short-term deflection of un-cracked PSC beams is calculated by elastic theory with the gross (or transformed) $E_c I$: $d^2y/dx^2 = M/EI$.

Beam & load	Max. deflection
Simply supported, UDL w	$\delta = 5\,w\,L^4 / (384\,E\,I)$
Simply supported, central point load W	$\delta = W\,L^3 / (48\,E\,I)$
Simply supported, constant moment M	$\delta = M\,L^2 / (8\,E\,I)$
Cantilever, UDL w	$\delta = w\,L^4 / (8\,E\,I)$
Cantilever, end load W	$\delta = W\,L^3 / (3\,E\,I)$
Cantilever, end moment M	$\delta = M\,L^2 / (2\,E\,I)$

Moment-area method (for any tendon profile)

- Deflection at mid-span = moment of the M/EI diagram between the support and mid-span, taken about the support (for symmetrical loading) / tangential deviation.
- Prestress moment at section = $P \cdot e_x$ (e_x = eccentricity at x). The curvature diagram $M/EI = P\,e_x / EI$.
- Prestress M-diagram for parabolic tendon = parabola; for bent tendon = triangle; straight = rectangle.



M-diagrams of prestress (shape ~ tendon profile)

Fig. Bending-moment diagrams from different tendon profiles

Sign convention : **upward deflection (camber) +ve** for prestress effects, downward for load effects – subtract to get net.

4. Deflection due to Prestress (Short-term, Un-cracked)

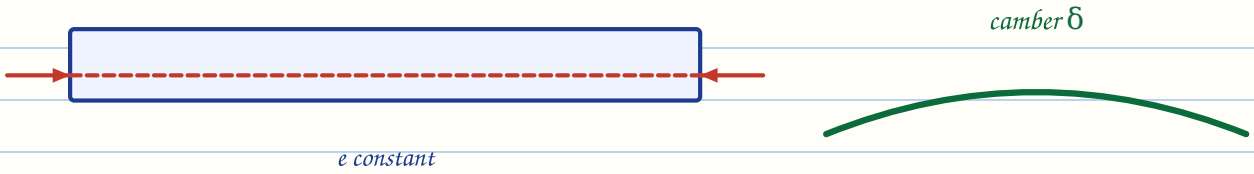
(a) Concentric straight tendon

$e = 0 \rightarrow$ no moment $\rightarrow$ **no deflection** (only shortening).

(b) Eccentric straight tendon (constant e)

Constant moment $M = P\,e$. Curvature $= Pe/EI$ (constant) $\rightarrow$ circular arc.

$$\delta = P\,e\,L^2/(8\,E\,I) \text{ (upward at mid-span)}$$



(c) Bent (harped) tendon – single kink at mid-span, e at mid, 0 at ends

- Equivalent upward load at mid-span $W = 4\,P\,e\,/\,L$; $\delta = W\,L^3/48EI$:

$$\delta = P\,e\,L^2/(12\,E\,I)$$

- For a **two-point bent tendon** (kinks at distance a from each support, e at the kinks) : each kink carries $W = P\,e/a$ upward and $\delta = P\,e\,(3L^2 - 4a^2)/(24\,E\,I)$ ($a = L/2$ gives back $PeL^2/12EI$ ✓).

(d) Parabolic tendon – e at mid-span, 0 at the ends

- Equivalent upward UDL $w_p = 8Pe/L^2$; $\delta = 5\,w_p\,L^4/384EI$:

$$\delta = 5\,P\,e\,L^2/(48\,E\,I)$$

(e) Parabolic tendon with eccentricity e_a at ends and e_c at mid-span

$$\delta = P\,L^2\,(e_a + 5\,e_c)/(48\,E\,I) = P\,e_a\,L^2/8EI + 5P(e_c - e_a)\,L^2/48EI$$

(e below C.G. +ve; e above C.G. –ve. For $e_a = 0 \rightarrow 5PeL^2/48EI$ ✓ ; for $e_a = e_c \rightarrow PeL^2/8EI$ ✓.)

Cantilever : straight tendon : $\delta = P\,e\,L^2/(2EI)\uparrow$; parabolic tendon (e at the fixed end, horizontal tangent at the free end) : $\delta = P\,e\,L^2/(12EI)\uparrow$. In every case integrate M/EI to get the coefficient.

5. Worked Examples (Short-term)

Example 1 – straight eccentric tendon

A PSC beam 150 × 300 mm, span 8 m, P = 300 kN at e = 50 mm (constant). E_c = 35000 N/mm². Self-weight + imposed load = 1.125 + 3 kN/m. Find camber, deflection and net deflection. Loss 15 %.

$$I = 150\times300^3/12 = 3.375\times10^8\ mm^4\ ;\ E_cI = 1.181\times10^{13}\ N\cdot mm^2$$

$$(i)\ Prestress: \delta_p = PeL^2/8EI = 300\times10^3\times50\times8000^2/(8\times35000\times3.375\times10^8) = 10.16\ mm\uparrow$$

$$(ii)\ Self\ weight: \delta_g = 5\times1.125\times8000^4/(384\times35000\times3.375\times10^8) = 5.08\ mm\downarrow$$

$$(iii)\ At\ transfer: net = 10.16 - 5.08 = 5.08\ mm\uparrow\ (camber)$$

$$(iv)\ Imposed\ 3\ kN/m: \delta_q = 13.54\ mm\downarrow;\ P_e = 0.85P \rightarrow \delta_p = 8.64\ mm\uparrow$$

$$(v)\ At\ service: net = 5.08 + 13.54 - 8.64 = 9.98\ mm\downarrow < L/250 = 32\ mm\ \checkmark$$

Example 2 – bent tendon

Beam 200 × 400, span 6 m, bent tendon with e = 80 mm at mid-span (0 at the ends), P = 500 kN, E_c = 36000.

$$I = 200\times400^3/12 = 1.0667\times10^9\ mm^4$$

$$\delta = PeL^2/(12EI) = 500\times10^3\times80\times6000^2/(12\times36000\times1.0667\times10^9) = 3.13\ mm\uparrow$$

Equivalent load $W = 4Pe/L = 4\times500\times0.08/6 = 26.7$ kN upward at mid-span ; check $\delta = WL^3/48EI = 26.7\times10^3\times6000^3/(48\times36000\times1.0667\times10^9) = 3.13$ mm ✓.

Example 3 – parabolic tendon, eccentric at the ends

Beam 300 × 600, span 12 m, M50 (E_c = 40305), P = 1000 kN. Tendon parabolic, 200 mm **above** C.G. at the ends and 200 mm **below** at mid-span. Find the camber.

$$I = 5.4\times10^9\ ;\ e_a = -200,\ e_c = +200$$

$$\delta = PL^2(e_a + 5e_c)/(48EI) = 10^6\times(12000)^2\times(-200 + 1000)/(48\times40305\times5.4\times10^9) = 11.0\ mm\uparrow$$

6. Example 4 – Complete deflection check (Design Ex. of Unit III)

Beam 300 × 500 mm, span 10 m, parabolic tendon (e = 120 mm at mid-span, 0 at supports), P_i = 1057 kN, losses 20 % (η = 0.8), self weight 3.75 kN/m, imposed 10 kN/m, M40 → $E_c = 5700\sqrt{40} = 36050$ N/mm². Find the short-term deflections at transfer and service.

$$I = 300\times500^3/12 = 3.125\times10^9 \text{ ; } E_cI = 1.1266\times10^{14} \text{ N·mm}^2$$

At transfer (P_i + self weight)

$$\delta_p = 5PeL^2/48EI = 5\times1057\times10^3\times120\times10^8/(48\times1.1266\times10^{14}) = 11.73 \text{ mm} \uparrow$$

$$\delta_g = 5wL^4/384EI = 5\times3.75\times10^{16}/(384\times1.1266\times10^{14}) = 4.33 \text{ mm} \downarrow$$

Net at transfer = 11.73 − 4.33 = **7.40 mm ↑** (camber ≈ L/1350 ✓ small)

At service (ηP_i + all loads, 13.75 kN/m)

$$\delta_p = 0.8\times11.73 = 9.38 \text{ mm} \uparrow \text{ ; } \delta_w = 4.33\times13.75/3.75 = 15.88 \text{ mm} \downarrow$$

Net = 15.88 − 9.38 = **6.50 mm ↓** < L/250 = 40 mm ✓

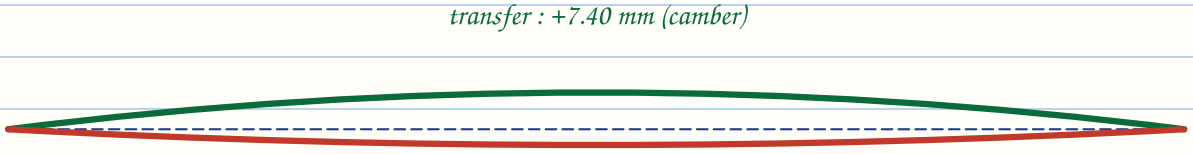


Fig. Net deflection profile at transfer and at service

Check of un-cracked behaviour (cracking moment)

- Section is un-cracked if no tension / tension < modulus of rupture $f_{cr} = 0.7\sqrt{f_{ck}} = 4.43$ N/mm².
- Cracking moment $M_{cr} = (f_{cr} + \eta P_i/A + \eta P_i e/Z)Z = (4.43 + 5.64 + 8.12)\times12.5\times10^6 = 227$ kN·m > M_T = 171.9 kN·m ✓ → un-cracked ⇒ gross I valid.

7. Cracked Sections – A Brief Note

When $M > M_{cr}$ the section cracks, the effective I reduces and deflections increase rapidly (stiffness of a Type 3 / partially prestressed member). Our syllabus is for **un-cracked beams**, but one should know:

- **Bilinear (M-δ) behaviour** – before cracking: gross I ; after cracking: cracked-transformed I_{cr}.
- **Effective moment of inertia (Branson)** : $I_{eff} = I_{cr} + (I_g - I_{cr})(M_{cr}/M_a)^3 \le I_g$.
- Deflection then : δ = (deflection by I_{eff}) using net moment $M_{net} = M_T - P\,e$.

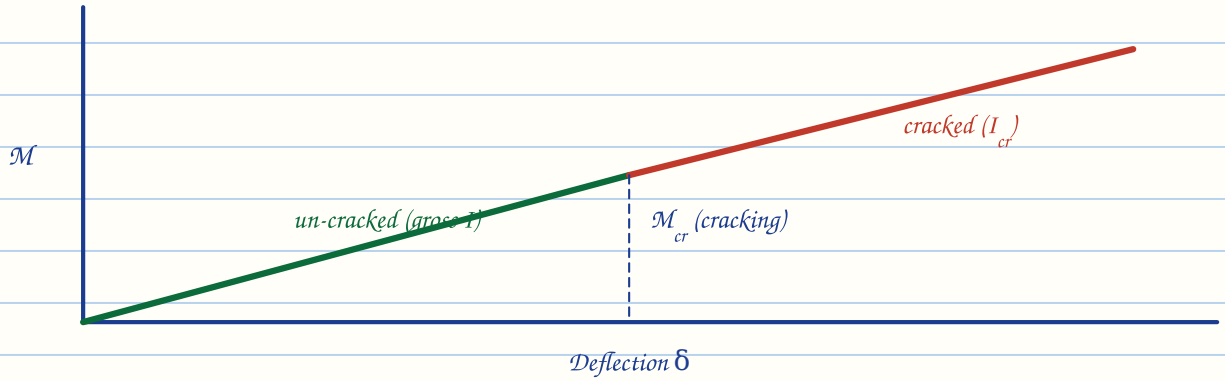


Fig. Moment – deflection curve of a prestressed beam

Transformed section

For accuracy, use the transformed area $A_t = A + (m - 1)\,A_{ps}$ and I_t – the correction is small (≈ 2–5 %). Normal practice is to use the gross section for deflection of un-cracked beams.

Remember : $M_{cr} = Z(f_{cr} + P_e/A) + P_e\,e$ (for the bottom fibre) ; $f_{cr} = 0.7\sqrt{f_{ck}}$ (IS 1343).

Prestressing delays cracking – the member remains stiff (gross I) over the full working-load range ⇒ much **smaller deflections than RCC** of equal span/depth.

8. Prediction of Long-Time Deflections

Under sustained loads concrete creeps and shrinks, steel relaxes and prestress decreases. Deflection and camber change with time. **Long-term deflection ≠ short-term.**

Effects to consider

1. **Creep of concrete** – increases curvature under sustained moment (both load and prestress effects).
2. **Loss of prestress** – reduces the upward camber with time.
3. **Shrinkage** – shortens concrete; with eccentric steel produces additional curvature.
4. **Increase in E_c** with age – reduces deflection slightly (usually neglected).

Method 1 – Effective modulus method (IS 1343 cl. 23.2)

$$E_{ce} = E_c / (1 + \theta)$$

- θ = creep coefficient (IS 1343 : 2.2 – 7 days ; 1.6 – 28 days ; 1.1 – 1 year).
- Long-term deflection = short-term deflection × $(1 + \theta)$, using the same P_e and sustained loads (P_e includes time-dependent losses).
- Net long-term $\delta_{LT} = (\delta_{loads} - \delta_{prestress})\,(1 + \theta)$ with $P = P_e$.

Method 2 – Multiplier method (PCI) for camber/deflection

Stage	Camber (due to P)	Deflection (due to self-weight etc.)
At erection (≈ 30–60 days)	× 1.80	× 1.85
Final (long-time)	× 2.45	× 2.70

- Multipliers applied to the initial (transfer) elastic camber / deflection ; superimposed dead load × 3.0 (final).
- Net final deflection = $2.70\,\delta_g - 2.45\,\delta_p$ + (load not in the initial stage) × $(1 + \theta)$.

Method 3 – Rate of creep / step-by-step (outline)

Divide the time into intervals; compute curvature increments from creep ($\Delta\varepsilon = \theta\,\Delta f/E$), losses and shrinkage – used in computer programs (ACI 209 / CEB-FIP models).

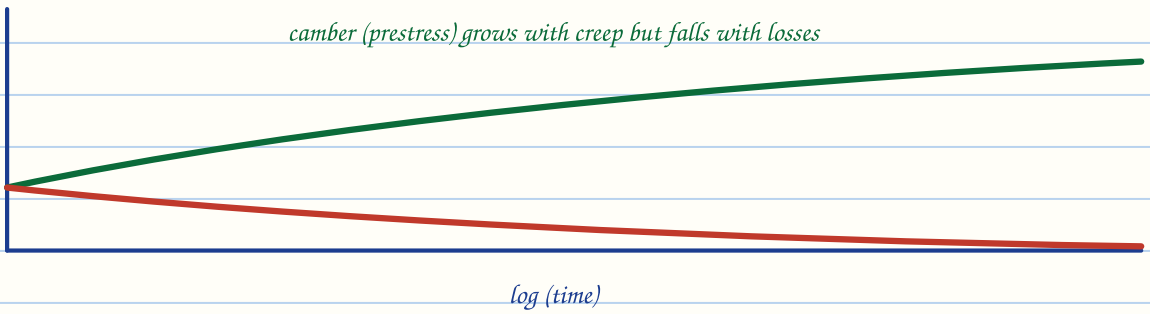


Fig. Time-dependent change of deflection

9. Worked Example – Long-term Deflection

Example 5

For Example 4 (300 × 500, L = 10 m, parabolic tendon, P_i = 1057 kN) compute the long-term deflection after creep and losses. θ = 1.6 (28 days), η = 0.8, all service loads sustained (3.75 + 10 = 13.75 kN/m).

Method 1 – Effective modulus

Short-term at service ($P_e = 0.8P_i$) : δ_w = 15.88 ↓ ; δ_p = 9.38 ↑ ; net = 6.50 mm ↓

Long-term : $\delta_{LT} = (1 + \theta) \times$ net = 2.6 × 6.50 = 16.9 mm ↓

Check : 16.9 mm < $L/250$ = 40 mm ✓ ; $L/350$ = 28.6 mm (after finishes) ✓

Method 2 – PCI multipliers

Initial (transfer) : δ_g = 4.33 ↓ , δ_p = 11.73 ↑ (before losses).

Final = 2.70 × 4.33 – 2.45 × 11.73 + (superimposed 10 kN/m : 11.55 mm) × 3.0 …

= 11.69 – 28.74 + 34.65 = 17.6 mm ↓

Both methods agree (≈ 17 mm ↓) – O.K. Camber at transfer was 7.4 mm ↑ (growing to ≈ 17.1 mm ↑ under self-weight + P at erection × multipliers).

Shrinkage curvature (unsymmetrical steel) : $\kappa_{sh} = \varepsilon_{cs} \cdot m \cdot A_{ps} \cdot e / I_t$; $\delta_{sh} = \kappa_{sh} L^2/8$ (SS beam). For this beam : $\kappa = 3\times10^{-4}\times5.8\times987\times120/3.125\times10^{9} = 6.6\times10^{-8}$/mm → δ = 0.83 mm ↓ (small ✓ ; included roughly in $(1 + \theta)$).

Long-term deflection is typically **2 – 3 times** the short-term deflection ; camber can **reverse** into downward deflection with time if load is heavy.

10. Control of Deflections

Design measures

Measure	How it helps
Increase depth D (span/depth ≤ 20–30)	$I \propto D^3 \rightarrow$ deflection ↓ strongly
Use **high-grade concrete**	↑ E_c, ↓ creep & shrinkage
Adopt **full / limited prestress** (un-cracked)	Gross I valid, camber helps
Adjust **P and e** (profile) properly	Camber balances dead load (load balancing)
Use **parabolic / harped** tendon with high e at mid-span	Larger camber, lower end stresses
Reduce losses : low-relaxation steel, steam curing, good curing	Retains camber/prestress
Provide **precamber** in formwork / compression steel	Offsets long-term deflection
Continuity / composite action / T-section	↑ stiffness, ↓ moment
Use **non-prestressed** compression steel	Reduces creep curvature
Delay application of finishes, partitions; propping	Reduces deflection after fixing finishes
Limit transfer-time camber : control f_{ci}, release	Avoid excessive camber (top cracking)

Span / effective-depth ratios (practical guide)

Member	L/D
Simply supported beams (PSC)	20 – 30
Continuous beams	25 – 35
Slabs	30 – 45
Cantilevers	8 – 12

Deflection of a PSC beam is far less than RCC – but because the members are slender, **always check at transfer and long-term.**

11. Important Questions & Formula Sheet

Short answers

1. State the IS 1343 limits for deflection.
2. List the factors influencing the deflection of a PSC beam.
3. Write the camber for parabolic tendon with end eccentricity.
4. What are the effects of creep and shrinkage on deflection ?
5. What is the effective modulus of elasticity ?
6. Differentiate un-cracked and cracked deflection behaviour.

Long answers

1. Derive the deflection of a simply supported beam with (a) straight eccentric (b) bent (c) parabolic tendon.
2. A PSC beam … calculate the short-term deflection at transfer and at service (numerical).
3. Explain the methods of predicting the long-time deflection of PSC beams. Numerical using $(1 + \theta)$.
4. Describe the measures to control deflection of PSC members.

Formula sheet

Item	Formula
Straight eccentric	$\delta = P e L^2 / 8EI \uparrow$
Bent (mid)	$\delta = P e L^2 / 12EI \uparrow$
Parabolic	$\delta = 5 P e L^2 / 48EI \uparrow$
Parabolic with end e_a	$\delta = P L^2 (e_a + 5e_c) / 48EI$
UDL	$\delta = 5 w L^4 / 384EI \downarrow$
Central point load	$\delta = W L^3 / 48EI \downarrow$
Long-term (IS)	$\delta_{LT} = (1 + \theta)\delta$; $E_{ce} = E_c/(1+\theta)$
PCI multipliers	1.80 / 1.85 (erection) ; 2.45 / 2.70 (final)
IS limits	$L/250$; $L/350$ or 20 mm
E_c	$5700\sqrt{f_{ck}}$ N/mm²

All the best ! Practise at least 2 numericals each for short-term and long-term deflection.

PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – V

COMPOSITE BEAMS

Syllabus : Composite beams – different types – propped and un-propped – stress distribution – differential shrinkage – analysis of composite beams.

Contents

1. Introduction & advantages of composite construction
2. Types of composite beams
3. Propped & un-propped construction
4. Stages of loading and stress analysis
5. Properties of composite section (effective width)
6. Worked example – un-propped construction
7. Worked example – propped construction
8. Differential shrinkage – theory
9. Differential shrinkage – example
10. Horizontal (interface) shear & shear connectors
11. Ultimate strength of composite beams
12. Important questions & formula sheet

Exam tip : The standard question is "analyse the composite beam for the three stages and find the stresses" – draw the stress diagrams for each stage and add them.

1. Introduction to Composite Construction

A **composite beam** consists of a **precast prestressed beam** and a **cast-in-situ concrete slab (or topping)** acting together as one unit through bond / shear connectors (stirrups projecting from the beam).

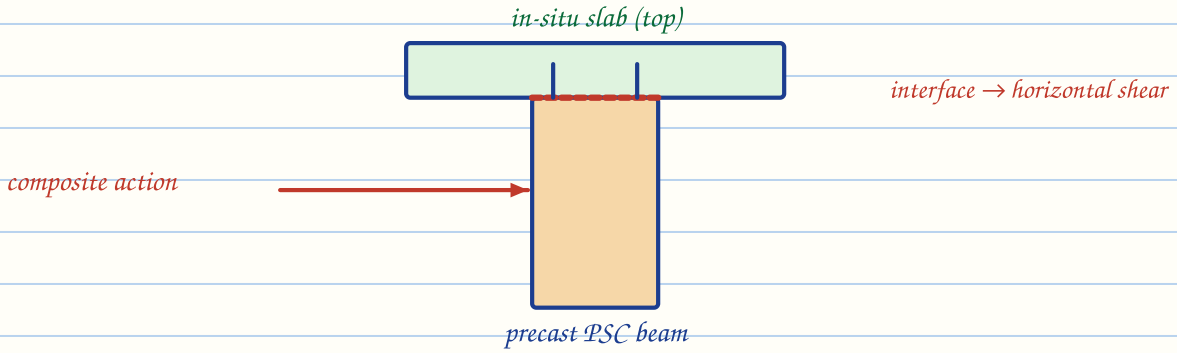


Fig. Composite section – precast beam + in-situ slab (shear ties across interface)

Why composite ?

- Precast beams are made in factory (pre-/post-tensioned) – good quality, quick erection; slab cast in-situ over them gives the deck (bridges, buildings, floors).
- The slab acts as a **compression flange** – increases the section modulus and moment of resistance → lesser prestress force and steel.
- Precast beam acts as the **formwork / support** for the slab (saving in shuttering – especially in bridges).
- Continuity of the deck can be provided by the in-situ concrete; monolithic appearance.
- Speed of construction, economy – the part of load (self-weight) is carried by the precast section alone.

Limitations

- Differential shrinkage and creep between precast and in-situ concrete causes additional stresses.
- Interface shear has to be provided for carefully (roughening, stirrups).
- The slab concrete is of lower grade (M25–M30) than the precast beam (M40–M60).
- Composite action starts only after the in-situ concrete has hardened – **un-propped** beams carry slab weight alone.

2. Types of Composite Beams

Type	Description
1. Precast PSC beam + cast-in-situ RC slab	Most common (T-beam bridge / floor). Beam pre-/post-tensioned; slab 100–200 mm.
2. Precast PSC beam + precast slab (with in-situ joints)	Fully precast deck, composite after grouting/screeding.
3. Precast RC/PSC beam + in-situ topping on hollow-core / ribbed slabs	Floors in buildings; topping 40–75 mm.
4. In-situ PSC + in-situ slab (two-stage casting)	Beam prestressed first; slab cast afterwards; long span.
5. Steel beam + concrete slab (steel–concrete composite)	Not prestressed generally; also possible with prestressing.
6. Post-tensioned precast + continuity (stitching)	Precast girders made continuous with in-situ crosshead / deck prestressing.

Common cross-sections

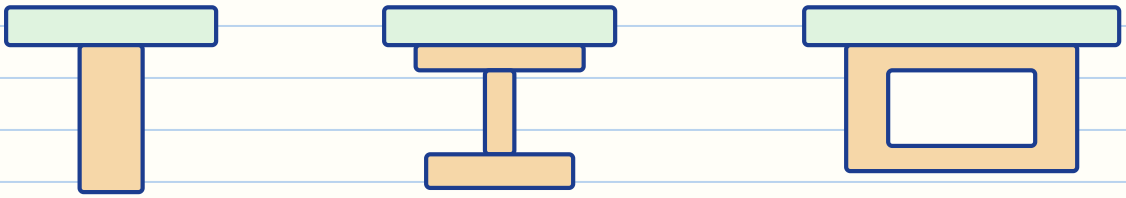


Fig. Typical composite sections (green = in-situ slab)

Based on construction method

- **Un-propped** – the precast beam carries its own weight + wet in-situ slab + formwork alone; composite action for later loads only.
- **Propped** – the beam is supported by props (temporary supports) while the in-situ slab is cast; the props are removed after the slab gains strength → the **composite section carries the whole weight**.

3. Propped vs Un-propped Construction

Aspect	Un-propped	Propped
Slab weight carried by	Precast beam alone (section 1)	Composite section (after props removed)
Props	None – simple, cheaper, faster	Props / shores required – costly, slow
Precast beam stress under slab weight	High (beam bends under wet slab)	Low – beam only carries own weight
Section needed	Stronger / deeper precast beam	Lighter precast beam possible
Total prestress required	Larger	Smaller
Deflection	Larger (before composite action)	Smaller
Suitability	Bridges over rivers/roads, high clearance	Buildings, low heights, short spans
Final stress in beam bottom	Lower compressive (more tension)	More favourable

(a) UN-PROPPED : slab load W_s on precast beam only



(b) PROPPED : props carry slab load until it hardens



Fig. Un-propped and propped construction

Un-propped : Stage 2 loads (slab weight) act on the **precast section**. Propped : slab weight acts on the **composite section** when the props are removed.

4. Stages of Loading and Stress Distribution

Stress at any fibre = sum of the stresses of all stages, each computed with the **section that actually resists it**.

Stage	Loads	Section resisting	Prestress
1. Transfer	$P_i + M_{g1}$ (beam self-weight)	Precast beam (A_1, I_1)	P_i
2. Casting of slab (un-propped)	$+ M_s$ (wet slab, formwork)	Precast beam alone	P_e (after losses)
3. After slab hardened	$+ M_q$ (finishes, live load, super-imposed)	**Composite** section (A_c, I_c)	P_e

Stress equations (compression +ve)

$$\text{Stage 1/2 (precast, unpropped)}: f = P_e/A_1 \mp P_e e/Z_1 \pm (M_{g1} + M_s)/Z_1$$

$$\text{Stage 3 (composite)}: f = \pm M_q / Z_c \text{ (at the level considered)}$$

Final stress = Stage 2 stress + Stage 3 stress (superposition)

- **At the interface**, the precast beam top fibre and slab bottom fibre have different stage stresses; but the strain is continuous after composite action (for the same E_c).
- The slab carries only the **stage-3** stress (and shrinkage effects) – none of the prestress.
- If E_c of slab ≠ E_c of beam, use modular ratio $m = E_{slab}/E_{beam}$ to transform the slab width : $b_{transformed} = m \cdot b_f$.

Effective flange width (IS 456 cl. 23.1.2)

$$b_f = L_0/6 + b_w + 6\,D_f \text{ (T-beam)}\ ;\ \le \text{spacing of beams}$$

L_0 = distance between points of zero moment (= span for simply supported beam), b_w = web width, D_f = slab thickness.

5. Worked Example – Un-propped Beam (data)

A precast prestressed rectangular beam 250 mm × 500 mm, simply supported, span 10 m, supports an in-situ slab 1000 mm × 100 mm (beam spacing 1 m; the code effective width $L/6 + b_w + 6D_f$ = 2517 mm > 1000, so 1000 mm governs). Construction un-propped. Same grade of concrete (m = 1).

Data	Value
Prestress after transfer P_i	700 kN at e = 130 mm below C.G. of the precast beam
Effective prestress $P_e = 0.8\ P_i$	560 kN
Self-weight of beam	0.25 × 0.5 × 25 = 3.125 kN/m
Weight of wet slab	1.0 × 0.1 × 25 = 2.5 kN/m
Imposed (finishes + live) on composite	7 kN/m

Moments (L = 10 m, $wL^2/8 = 12.5\ w$)

$M_{g1} = 3.125 \times 12.5 = 39.06$ kN·m ; $M_s = 2.5 \times 12.5 = 31.25$ kN·m ; $M_q = 7 \times 12.5 = 87.5$ kN·m

Section properties

Precast beam alone (section 1) :

$A_1 = 125000\ mm^2$; $I_1 = 250 \times 500^3/12 = 2.604 \times 10^9\ mm^4$; $Z_1 = I_1/250 = 1.0417 \times 10^7\ mm^3$

Composite section (section 2) – slab 1000 × 100 on top of beam :

$A_c = 125000 + 100000 = 225000\ mm^2$

$\bar{y}_b = (125000 \times 250 + 100000 \times 550)/225000 = 383.3$ mm above the beam bottom ; $\bar{y}_t = 600 - 383.3 = 216.7$ mm below slab top

$I_c = 2.604 \times 10^9 + 125000 \times 133.3^2 + 8.333 \times 10^7 + 100000 \times 166.7^2 = 7.687 \times 10^9\ mm^4$

y at beam bottom = 383.3 ; at interface (beam top) = 116.7 above C.G. ; at slab top = 216.7 above C.G.

6. Stress Analysis – Un-propped

Stage 1 – at transfer (precast alone, $P_i + M_{g1}$)

$$P_i/A_1 = 5.60 \; ; \; P_i e/Z_1 = 700\times10^3\times130/1.0417\times10^7 = 8.74 \; ; \; M_{g1}/Z_1 = 3.75 \; (N/mm^2)$$

$$\text{Top} = 5.60 - 8.74 + 3.75 = +0.61 \; ; \; \text{Bottom} = 5.60 + 8.74 - 3.75 = 10.59 \; ✓$$

Stage 2 – slab cast (precast alone, $P_e + M_{g1} + M_s$)

$$P_e/A_1 = 4.48 \; ; \; P_e e/Z_1 = 6.99 \; ; \; (M_{g1} + M_s)/Z_1 = 70.31\times10^6/1.0417\times10^7 = 6.75$$

$$\text{Top (beam)} = 4.48 - 6.99 + 6.75 = 4.24 \; ; \; \text{Bottom} = 4.48 + 6.99 - 6.75 = 4.72$$

Stage 3 – imposed load on composite section ($M_q = 87.5$ kN·m)

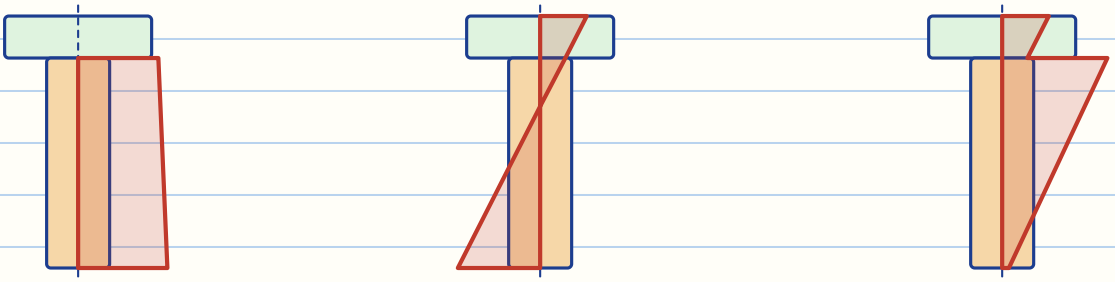
Fibre	y from C.G. (mm)	$M_q y/I_c$ (N/mm²)
Slab top	216.7 ↑	$87.5\times10^6\times216.7/7.687\times10^9$ = **+2.47 C**
Slab bottom / beam top (interface)	116.7 ↑	**+1.33 C**
Beam bottom	383.3 ↓	**−4.36 T**

Final stresses (N/mm²)

Fibre	Stage 2	Stage 3	Final
Slab top	–	+2.47	**2.47 C**
Slab bottom (interface)	–	+1.33	**1.33 C**
Beam top (interface)	+4.24	+1.33	**5.57 C**
Beam bottom	+4.72	−4.36	**0.36 C**

Beam bottom is still in compression (0.36 N/mm²) – **no tension** ✓. Permissible compression at beam top 5.57 ≪ 16 N/mm² ✓.

Stress diagrams (un-propped)



Stage 2 : $P_e + M_{g1} + M_s$ (beam only) | Stage 3 : M_q (composite) (composite section) | Final (sum of stages)

Fig. Stress distribution – stage-wise and final (red area = stress, right = compression)

- In stage 3 the stress diagram is **one straight line** over the whole composite depth (plane sections remain plane).
- Final diagram has a **jump** at the interface – the slab does not carry prestress / the stage-2 stresses.

7. Worked Example – Propped Construction (same data)

Props carry the slab weight → precast beam alone carries $P_e + M_{g1}$. After removing the props, the composite section carries $M_s + M_q = 31.25 + 87.5 = 118.75$ kN·m.

Stage 2 (beam alone, $P_e + M_{g1}$) : Top = 4.48 – 6.99 + 3.75 = 1.24 ; Bottom = 4.48 + 6.99 – 3.75 = 7.72

Stage 3 (composite, $M = 118.75$) : $M/I_c = 118.75\times10^6 / 7.687\times10^8 = 0.01545$ N/mm³ ; stress = $0.01545 \times y$ →

Fibre	Stage 2	Stage 3	Final
Slab top	–	0.01545 × 216.7 = +3.35	**3.35 C**
Slab bottom	–	+1.80 (y = 116.7)	**1.80 C**
Beam top	+1.24	+1.80	**3.04 C**
Beam bottom	+7.72	−5.92 (y = 383.3)	**1.80 C**

Propped : beam bottom = 1.80 N/mm² (more compression) vs 0.36 N/mm² un-propped → propping improves the stress state and allows lesser P.

8. Differential Shrinkage

The precast beam is old (its shrinkage is mostly complete), whereas the in-situ slab is new and shrinks a lot after casting. The beam **restrains** the slab → stresses develop in the composite beam.

- The slab tends to shrink more → tensile stress in the slab (restrained), compression + tension in beam; a **sagging curvature** (deflection) is produced.
- **Differential shrinkage strain** $\varepsilon_{ds} = \varepsilon_{cs}$ (slab) $- \varepsilon_{cs}$ (precast) $\approx 1.5\times10^{-4}$ to 3×10^{-4} (IS 1343 : shrinkage strain 0.0003 pre – 0.0002/log(t+2) post).
- Creep of concrete **relieves** part of the stress – use the effective modulus $E_{ce} = E_c/(1+\theta)$ for the slab.

Analysis procedure (3 steps)

1. **Free shrinkage** : the slab is allowed to shrink freely – no stress.
2. **Restrain** the slab : apply a tensile force at the slab C.G. to bring it back to the original length : $N = \varepsilon_{ds} \cdot E_{ce} \cdot A_s$ (A_s = slab area) → tensile stress in slab = $N/A_s = \varepsilon_{ds} E_{ce}$.
3. **Release** the restraint : apply an equal and opposite force N (compressive) at the slab C.G. to the **composite section** – it acts at eccentricity e_s from the composite centroid.

$$\text{Stress from release : } f = N/A_c \pm N e_s y / I_c$$

$$\text{Final slab stress} = -N/A_s \text{ (restraint tension)} + N/A_c + N e_s y/I_c$$

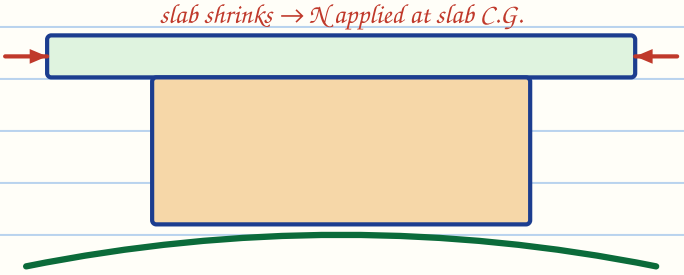


Fig. Differential shrinkage – restraint force and curvature

9. Differential Shrinkage – Example

For the composite beam of Example 5 : slab 1000 × 100, beam 250 × 500, E_c = 36000 N/mm², differential shrinkage strain $\varepsilon_{ds} = 2.0\times10^{-4}$, creep coefficient θ = 1.6. Find the stresses due to differential shrinkage.

$$E_{ce} = 36000/(1 + 1.6) = 13846\ N/mm^2$$

$$N = \varepsilon_{ds} E_{ce} A_s = 2\times10^{-4} \times 13846 \times 100000 = 276.9\ kN \text{ (slab tension if restrained} = 2.77\ N/mm^2)$$

$$e_s = 550 - 383.3 = 166.7\ mm\ ;\ N e_s = 276.9\times10^3 \times 166.7 = 4.615\times10^7\ N{\cdot}mm$$

$$N/A_c = 276900/225000 = 1.23\ N/mm^2$$

Fibre	N/A_c	$N e_s y/I_c$	Release stress	With slab restraint (−2.77)
Slab top	+1.23	+1.30 (y = 216.7 ↑)	+2.53 C	−0.24 (T)
Slab bottom	+1.23	+0.70	+1.93 C	−0.84 (T)
Beam top	+1.23	+0.70	+1.93 C	–
Beam bottom	+1.23	−2.30 (y = 383.3 ↓)	**−1.07 (T)**	–

- *The slab remains nearly stress free; the beam bottom receives a **tension of 1.07 N/mm²** which is subtracted from the earlier 0.36 C : final = 0.36 − 1.07 = **−0.71 N/mm² (tension)** → small tension; acceptable for Type 2 (limited tension) but not for Type 1 – design must provide more prestress or use propping.*
- *Curvature due to shrinkage $\kappa = N e_s/(E_c I_c)$ (with E_{ce}) → extra deflection $\delta = \kappa L^2/8$.*

$$\kappa = 4.615\times10^7/(13846\times7.687\times10^9) = 4.3\times10^{-7}/mm\ ;\ \delta = \kappa L^2/8 = 4.3\times10^{-7}\times10^8/8 = 5.4\ mm \downarrow$$

Always add the differential shrinkage stresses to the un-propped / propped stage stresses before checking the permissible limits.

10. Horizontal Shear at the Interface

The slab and the beam must act together – the **horizontal shear** along the interface is resisted by bond, friction of the roughened surface and **shear connectors (stirrups projecting into the slab)**.

$$\tau_h = V A_s \bar{y}_s / (I_c\, b)$$

- V = shear force acting on the composite section (after composite action, i.e. stage-3 loads : superimposed dead + live) ; A_s = area of the slab ; $\bar{y}_s$ = distance of slab C.G. from composite C.G.
- b = width of the interface (top width of the precast beam).

Example 6

At the support of the example beam $V = 7 \times 5 = 35$ kN (imposed) + live load, say total $V = 60$ kN on the composite section.

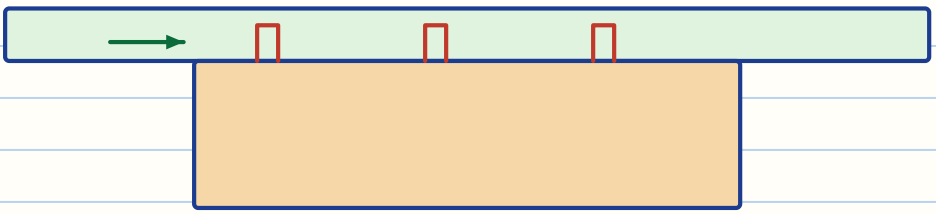
$$A_s \bar{y}_s = 100000 \times 166.7 = 1.667 \times 10^7 \text{ mm}^3$$

$$\tau_h = 60 \times 10^3 \times 1.667 \times 10^7 / (7.687 \times 10^9 \times 250) = 0.52 \text{ N/mm}^2$$

Compare with the permissible values (IS 456 / IS 1343 / IRC) for interface condition (roughened, with / without ties) – if higher, provide **vertical ties** : $A_v = \tau_h\, b\, s / (0.87\, f_y\, \mu)$.

Methods of providing composite action

- Roughen the top surface of the precast beam (≈ 5 mm amplitude), clean, wet.
- Projecting **stirrups** (shear keys) from the beam into the slab – spacing ≤ 300 mm / 4 × slab depth.
- Shear keys, indents, dowels, welded studs (steel–concrete composite).
- Adequate cover and anchorage of the ties in the slab and in the beam.



projecting stirrups act as shear connectors across the interface

Fig. Shear connectors (ties) at the beam–slab interface

11. Ultimate Strength of Composite Beams

At ultimate, the composite section is analysed like a PSC / RC T-section – the slab acts as the compression flange (neutral axis usually in the slab). Strain in steel includes the effective prestress strain.

- Compression $C = 0.36\, f_{ck}\, b_f\, x_u$ (f_{ck} of the slab) ; tension $T = A_{ps}\, f_{pu}$; $\Sigma H = 0 \rightarrow x_u$.
- $M_u = T\,(d - 0.42\, x_u)$; check $M_u \geq 1.5\, M_{service}$ (IS 1343 load factor 1.5 for dead + live).
- Stress f_{pu} from IS 1343 (Fig. 3 / design curves), depends on $A_{ps}/(b\, d)$, f_{ck} and effective prestress.

Example 7

For the above beam : $A_{ps} = 636$ mm² (≈ 6 strands 12.7 mm ; $P_i = 700$ kN), depth to tendon from top of slab $d = 100 + 250 + 130 = 480$ mm, f_{ck} (slab) = 40, assume $f_{pu} \approx 1400$ N/mm² (from IS 1343 curves), $b_f = 1000$ mm.

$$T = 636 \times 1400 = 890\ \text{kN} ; C = 0.36 \times 40 \times 1000 \times x_u = 14400\, x_u$$

$$x_u = 890 \times 10^3 / 14400 = 61.8\ \text{mm} < D_f = 100\ \text{mm} \checkmark \text{ (N.A. in slab)}$$

$$M_u = 890 \times (480 - 0.42 \times 61.8) = 890 \times 454 = \mathbf{404\ kN{\cdot}m}$$

$$M_{service} = 39.06 + 31.25 + 87.5 = 157.8\ \text{kN{\cdot}m} ; 1.5 \times 157.8 = 236.7\ \text{kN{\cdot}m} < 404 \checkmark \text{ safe}$$

Note : In un-propped construction the precast beam carries stage-2 moment alone; but at ultimate the composite section is assumed to act as a whole (as the beam and slab are connected) provided the interface shear is resisted.

Check also : (i) shear (V_{co}, V_{cr} with composite section), (ii) interface shear, (iii) deflection (including differential shrinkage curvature), (iv) cracking/fatigue if required.

12. Important Questions & Formula Sheet

Short answers

1. What is a composite beam ? State its advantages.
2. Differentiate between propped and un-propped construction.
3. What is differential shrinkage ? How is it handled in analysis ?
4. Define effective flange width of a T-beam (IS 456).
5. How is the horizontal shear between precast beam and slab resisted ?
6. What is the effect of creep on differential shrinkage stresses ?

Long answers

1. Explain the stages of loading and the stress distribution in a composite beam constructed (a) un-propped (b) propped. Compare them.
2. A composite beam consists of a PSC beam … Find the stresses at all stages (numerical – as Example 5).
3. Derive the stresses due to differential shrinkage in a composite beam. Numerical (Example 6).
4. Explain the design of shear connectors; compute the interface shear stress.
5. Explain the types of composite beams with sketches.

Formula sheet

Item	Formula
Stage 2 (precast alone)	$f = P_e/A_1 \mp P_e e/Z_1 \pm (M_{g1} + M_s)/Z_1$
Stage 3 (composite)	$f = M_q\, y / I_c$
Composite section	$\bar{y} = \Sigma Ay/\Sigma A$; $I_c = \Sigma I + \Sigma A\, d^2$
Effective flange width	$b_f = L_0/6 + b_w + 6D_f$
Modular ratio	$m = E_{slab}/E_{beam}$
Differential shrinkage force	$N = \varepsilon_{ds}\, E_{ce}\, A_s$, $E_{ce} = E_c/(1+\theta)$
Shrinkage stress	$f = N/A_c \pm N e_s\, y/I_c$ (release) $- N/A_s$ (slab)
Interface shear	$\tau_h = V A_s \bar{y}_s/(I_c\, b)$
Ultimate moment	$M_u = A_{ps} f_{pu} (d - 0.42\, x_u)$

Best wishes for your exams ! Practise the stage-wise stress tables – they carry most of the marks.